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IEEE Guide for the Application and Interpretation of Frequency Response Analysis for Oil-Immersed Transformers

STANDARDS

IEEE Power and Energy Society

Developed by the
Transformers Committee

IEEE Std C57.149™-2024
(Revision of IEEE Std C57.149™-2012)

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Approved 26 September 2024

IEEE-SA Standards Board

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Abstract: The measurement of frequency response analysis (FRA) of oil-immersed power transformers is applicable in this guide. It is intended to provide the user with the requirements and specifications for instrumentation, procedures for performing the tests, techniques for analyzing the data, and recommendations for long-term storage of the data and results.

Keywords: admittance, attenuation, Bode plot, deviation, frequency domain, FRA, frequency response analysis, IEEE C57.149™, impedance, magnitude, phase angle, resonance, RLC network, transfer function

The Institute of Electrical and Electronics Engineers, Inc.
3 Park Avenue, New York, NY 10016-5997, USA

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PDF: ISBN 979-8-8557-1703-7 STD27635
Print: ISBN 979-8-8557-1704-4 STDPD27635

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Introduction

This introduction is not part of IEEE Std C57.149-2024, IEEE Guide for the Application and Interpretation of Frequency Response Analysis for Oil-Immersed Transformers.

Frequency Response Analysis (FRA) testing has gained popularity for assessing the mechanical integrity of oil-immersed transformers. Due to limited understanding and available information regarding FRA requirements and specifications for instrumentation, procedures for performing the tests, and analysis of results, the Performance Characteristics Subcommittee formed the Working Group PC57.149. The primary objective of the Working Group PC57.149 was to compile and validate FRA experiences and techniques to develop a FRA application and interpretation guide that would benefit the industry.

IEEE Std C57.149 was first published in 2012. This revision is an update to the guide to reflect the current state of the art and has been brought up to date with current IEEE styles. New clauses discussing the influence of grounding on testing and the diagnosis of floating shields have been added. A new table of test connections has also been added, as well as, several new frequency scans. Clauses on contact resistance and winding looseness due to transportation have been removed and the document has generally been updated.

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IEEE Guide for the Application and Interpretation of Frequency Response Analysis for Oil-Immersed Transformers

1. Overview

1.1 Scope

This guide is applicable to the measurement of Frequency Response Analysis (FRA) of an oil-immersed power transformer. The guide includes the requirements and specifications for instrumentation, procedures for performing the tests, techniques for analyzing the data, and recommendations for long-term storage of the data and results. This guide can be used in both field and factory applications.

1.2 Purpose

The purpose of this guide is to provide the user with information that will assist in making frequency response measurements and interpreting the results from these measurements. It provides guidance for all current methods employed in taking these measurements.

1.3 Word usage

The word *shall* indicates mandatory requirements strictly to be followed in order to conform to the standard and from which no deviation is permitted (*shall* equals *is required to*).[6,7]

The word *should* indicates that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required (*should* equals *is recommended that*).

The word *may* is used to indicate a course of action permissible within the limits of the standard (*may* equals *is permitted to*).

The word *can* is used for statements of possibility and capability, whether material, physical, or causal (*can* equals *is able to*).

[6]The use of the word *must* is deprecated and cannot be used when stating mandatory requirements; *must* is used only to describe unavoidable situations.
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IEEE Std C57.12.70™- 2020, IEEE Standard for Standard Terminal Markings and Connections for Distribution and Power Transformers.[8,9]

IEEE Std C57.12.80™-2002, IEEE Standard Terminology for Power and Distribution Transformers.

3. Acronyms and Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause. [10]

DETC	de-energized tap changer
FRA	frequency response analysis
LTC	load tap changer

4. FRA measurement overview

Frequency response analysis is a sensitive diagnostic technique for detecting changes in the electrical characteristics of power transformer windings. Such changes can result from various types of electrical or mechanical stresses (shipping damage, seismic forces, loss of clamping pressure, short circuit forces, insulation failure, etc.). The test is non-destructive and non-intrusive and can be used either as a stand-alone tool to detect winding damage, or as a diagnostic tool to pinpoint damages detected in other tests (e.g., insulation power factor, dissolved gas analysis, or short circuit impedance tests). FRA consists of measuring the admittance or impedance of the capacitive and inductive elements comprising the transformer windings. The measurement is performed over a wide range of frequencies and the results are compared with a reference "signature" or "fingerprint" of the winding to make a diagnosis.

The FRA measurement provides diagnostic information, in the form of a transfer function, related to the RLC network of the test specimen. The RLC network is integrally related to the physical geometry and construction of the test specimen.

Physical changes within the test specimen alter the RLC network, and in turn can alter the transfer function. The transfer function behavior can reveal a wide range of mechanical or electrical changes in the test specimen. Different transformer failure modes can have different effects on the network admittances, and may alter the transfer function. It is also possible that a particular failure mode may have no recognizable effect on the transfer function at all.

FRA can often detect gross transformer defects, as can other electrical tests. However, because of the sensitivity of the test, a primary benefit of FRA is the potential for detection of defects in the mechanical or electrical integrity of the transformer that are not apparent with other electrical tests.

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4.1 Use of FRA

Since the FRA test is used to detect mechanical movement or damage in a transformer. Mechanical movement refers to the actual movement of transformer parts (coils, core, leads, or accessories) with respect to each other or to ground in such a manner as to change the internal inductances or capacitances of the test specimen. It is most appropriately used after some event or condition that has the possibility of causing mechanical movement or electrical damage to the transformer assembly. Some of the typical scenarios where FRA measurements might be beneficial to include the following:

— Factory short-circuit testing
— Installation or relocation
— After a significant through-fault event
— As part of routine diagnostic measurement protocol
— After a transformer alarm (i.e., sudden pressure, gas detector, Buchholz)
— After a major change in online diagnostic condition (i.e., a sudden increase in combustible gas, etc.)
— After a change in electrical test conditions (i.e., a change in winding capacitance)
— System modeling purposes

There are two distinct categories for application of FRA measurement: baseline measurement and diagnostic measurement. In both cases, the procedures and precautions used to generate a good measurement are the same. However, there is a difference in motivation for the tests in each category.

4.2 FRA base line measurement

The base line FRA measurement may be done in either the factory or the field, and it provides information that can be used for some future need. The several distinct reasons to generate base line FRA measurements are as follows:

— To provide a standard of comparison for future diagnostic FRA measurements
— Transportation diagnostics prior to relocation and commissioning
— Required by customer specification
— Prior to short-circuit testing
— Quality assurance

Important factors to consider when performing baseline FRA measurements include determining the necessary tests and connections that might later be needed for diagnostic purposes, documentation of methods and connections, archiving data, verification of results, and repeatability of the methods and results. This guide provides assistance in each of these areas.

The test configuration can have an impact on the test results. It may be difficult to determine if these minor variations are due to differences in test configuration or some other physical change. Therefore, it is important to document the test configuration and connections for future test repeatability.

4.3 FRA diagnostic application

The several distinct reasons to generate diagnostic FRA measurements within a factory or field environment are as follows:

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— Verification that no damage occurred during a short circuit test
— Relocation and commissioning validation
— Post incident verification: lightning, external through-fault, internal short circuit, seismic event, etc.
— Routine diagnostic purposes
— Condition assessment of older transformers
— Evaluation of used or spare transformers
— Shipping and receiving

Important factors when performing diagnostic FRA measurements include matching the set up and instrumentation parameters used for the baseline measurements. When baseline data is not available, then data on duplicate transformers or other identical phases of a three-phase transformer may be used. Typical data from other transformers of the same type may also be helpful for comparison. Special methods or preparation may be needed in certain field applications due to aging of the equipment and connections, field applied treatment to bushings, modification to the transformer since the baseline measurements were made, or problems in making good ground connections due to field painted surfaces. This guide provides assistance in these areas.

4.4 Recommended FRA measurement test parameters

Test equipment should produce a frequency response measurement with the following characteristics:

— The test should be made over a wide range of frequencies, from 20 Hz up to at least 2 MHz., so as to be able to diagnose problems in the core, clamping structure, windings and interconnections.
— Both the Magnitude and Phase Angle of the measured transfer function should be possible to present. The FRA magnitude is the signal amplitude relationship between the reference (input, Vin) and measured (output, Vout) signals.It is often represented as decibels: $MAG(dB) = 20 \times \log10(Vout/Vin)$, and contains the impedance effect of the measurement system. The FRA phase angle is the shift of the response relative to that of the injected signal.
— The test should be made over a wide range of frequencies so as to be able to diagnose problems in the core, clamping structure, windings and interconnections.
— Successive measurements should have adequate resolution to give unambiguous diagnosis.

The test equipment should have the following attributes:

— Calibrated to an acceptable standard.
— The output power of the excitation source should provide adequate power over the entire frequency range to allow for consistent measurement of the transfer function across the frequency range.
— The test set should be capable of measuring sufficient dynamic range, over the frequency range in order to accommodate most transformer test objects.
— The test set should be capable of collecting a minimum of 200 measurements per decade, either spaced linearly or logarithmically.
— The test system (set and leads) should provide a known and constant characteristic impedance. The test set and lead characteristic impedances should be matched.
— A three lead system, signal, reference and test, should be used to reduce effect of leads in the measurement.

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— Test leads should be coaxial cables as close to the same length as possible and less then 30 m (100 ft) long. Shielded test leads should have the ability to be grounded at either end.

— Both the magnitude and phase angle of the measured transfer function should be presented.

5. Making an FRA measurement

5.1 Test procedures

As with any electrical test, making a frequency response measurement should be done in a safe and controlled manner irrespective of test location. Considerations for electrical safety in testing apply not only to personnel, but also to the transformer and test equipment. Prior to testing, involved personnel should discuss the test procedure and environment for confirming that the work to be performed and any safety precautions are clearly understood. Other safety aspects are covered in industry standards, company or local regulations and manufacturer's instruction manual.

5.2 Test environment preparation

The following should be performed before beginning an electrical test:

— Any transformer under test shall be completely isolated from any high voltage source or power system source.

— The transformer tank shall be grounded.

— All instrumentation shall be grounded appropriately for the specific test setup, and isolated from any high voltage source or power system source. Avoid subjecting the test instrument, test leads, or power supply to station wiring surges, and external interference, including transferred potentials.

— During the test, there shall be strict adherence to local safety regulations and guidelines.

5.3 Test object preparation

It is recommended that the configuration of the transformer be in as close to 'in service' configuration as possible.

All external bushing connections should be disconnected. This includes phase connections, neutral connections, stabilizing windings, and tertiary grounds.

Whenever possible, all test lead connections should be made directly to the bushing terminals. Any extra conductor length that is included in the test circuit path will influence the FRA test result. Short lengths of bus bar attached to the transformer will not appreciably influence the measurement, as long as the test leads are connected directly to the bushing terminals after the attached bus bar, so that the bus bar is not part of the test circuit.

In instances where it is impossible to connect directly to a transformer bushing, it is possible to perform frequency response measurements with a short section of bus bar attached. This will affect the results but may be acceptable as a test technique where it is impossible to exclude such short lengths from the circuit. Examples include rigid connections in confined workspaces. It is important to note the state of the transformer under test so as to provide a consistent method of testing. Where a transformer in the field has been tested previously with small lengths of bus bar attached, it should be tested in the same way subsequently, if a comparison to historical data is necessary. Analysis of results should take in to account possible variations that may be caused by connections and their supports. As a general guideline, external bus bar connections should be avoided.

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Special consideration shall be given to safety when testing a transformer without oil so that excessive voltages are not applied or induced in a combustible environment. The results of frequency response measurements differ as a consequence of removing the oil. Testing with oil is the most common and preferred method for frequency response analysis.

When a transformer is equipped with a de-energized tap changer, it is a decision for the transformer owners as to whether they wish to operate the de-energized tap changer.

If internal current transformers are present, they should be configured for in-service conditions. If oil pumps are present, it should be verified that these are not running during the FRA-measurement.

5.4 Test set

The test set should be grounded according to the recommendations of the test equipment manufacturer, or to the same point as the transformer under test, in the absence of equipment manufacturer's recommendations. Generally, the transformer tank ground should be considered as reference potential for the FRA measurement. It should be noted that in all FRA measurements, the grounding techniques will have a significant effect on test results. Grounding techniques, including selection of ground conductors as well as their routings, should therefore be precise, repeatable, and documented.

The test equipment should always be within the recommended calibration interval. When possible prior to use, a self-check of the operation of the test equipment using a standard test object with a known FRA response may be employed as a means of assuring correct operation of the equipment. This check is especially valuable for checking FRA test equipment, since there is generally no intuitive way of knowing if the test equipment is giving correct results when making field measurements.

5.5 Test leads

The following four coaxial leads should be used in addition to the shield grounds:

— Excitation "source"

— Specimen input "reference"

— Specimen output "measure"

— Measurement and reference cable grounds

These leads should be as close to the same length as possible and have a characteristic impedance matched to the test set. Ideally the leads should be the same length. As a minimum, the "reference" and "measure" leads should be identical.

Test leads should be checked for continuity and integrity before use. The best means for checking lead integrity is to perform the FRA self-check using a standard test object.

Where shorting leads are used as part of a test set up between bushing terminals, these should be insulated from ground, and be as short as possible. The impedance of these leads will influence the test results. Therefore, when the test procedure requires shorting of terminals, selection of shorting conductors as well as their routings should be precise, and repeatable, and documented.

Where local recommendations and/or guidelines require test grounds be applied to separate windings not under test, these grounds should be as short as possible and connected to the same ground as the transformer. It should be recognized that while the FRA response is not invalidated by the presence of additional winding grounds, the response with these grounds in place may be unique, and should not be compared with previous

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FRA test results obtained without the grounds installed. For the test to yield maximum value, every effort should be made to configure the test object exactly as recommended by the test equipment manufacturer. If necessary, requests may be made to the appropriate authority whenever it is deemed necessary to temporarily disconnect ground connections to separate windings, as long as the transformer is fully isolated from other power sources, and no hazards to safety are generated by the proximity of the transformer terminals to other energized substation equipment. In all cases, special permission should be received from the appropriate authority to deviate from any local recommendations and/or guidelines.

General lead connection diagrams are shown below in Figure 1, which provide examples of a typical test setup.

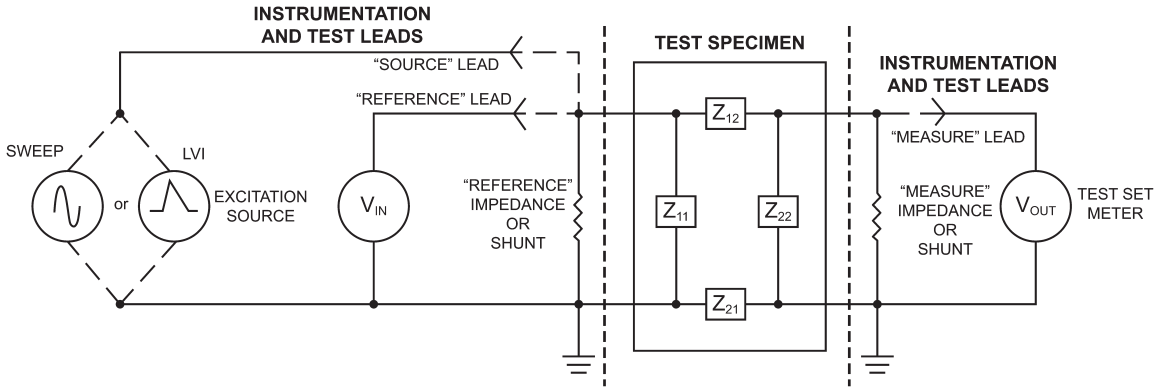


Figure 1—General lead connection diagram

These leads should be as close to the same length as possible and have a characteristic impedance matched to the test set. Ideally the leads should be the same length. As a minimum, the "reference" and "measure" leads should be identical and located as much as possible in similar positions. Test leads should be checked for continuity and integrity before use. The best means for checking lead integrity is to perform the FRA self-check using a standard test object.

The shields of the coaxial test leads should be connected to a common ground. This helps reduce the impedance between the shields that will influence the measurements at higher frequencies; typically differences between these methods can occur above 500 kHz. Common methods used include the following:

a) The method of using a shortest possible braid along the bushing surface to connect the shields to the bushing flange.

b) Using a fixed length wire, not coiled, connect the shields to the bushing flange, transformer tank ground.

c) Using a common, grounded, connection point near the bushings under test where the test leads shields are connected.

When connections are made to the bushing flange or mounting bolts, rust, paint and oxidation will create adverse conditions. The impedance of shield grounding will influence the test results. Therefore, their routings should be with aim to minimize impedance and be well documented.

5.6 Influence of grounding

The importance of good measurement lead connections cannot be overstated. Poor connections can result in measurement errors causing confusion and may bring the user to question the transformer integrity. This

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applies not only to the measurement leads, (source, reference, measurement etc.) but also the shield grounds of those leads. It is crucial that the grounding path not introduce any additional impedance into the circuit. To that end, care should be taken to help ensure the connections are solid and verified. Quite often when connections are made to the bushing flange or mounting bolts, rust, paint, and oxidation will create adverse conditions.

Ground problems can manifest themselves either through user error, lead placement, or poor connections. The following are common sources of ground-related problems:

— Source/reference lead ground problem
— Measurement lead ground problem
— All (source/reference and measurement) ground problems
— Bushing flange ground problem
— Test instrument overall ground problem

A poor test equipment ground does not significantly affect the measurement, however it is a critical component regarding safety; the test instrument chassis shall always be properly grounded.

Figure 2 and Figure 3 are examples of the responses with solid and poor grounding.

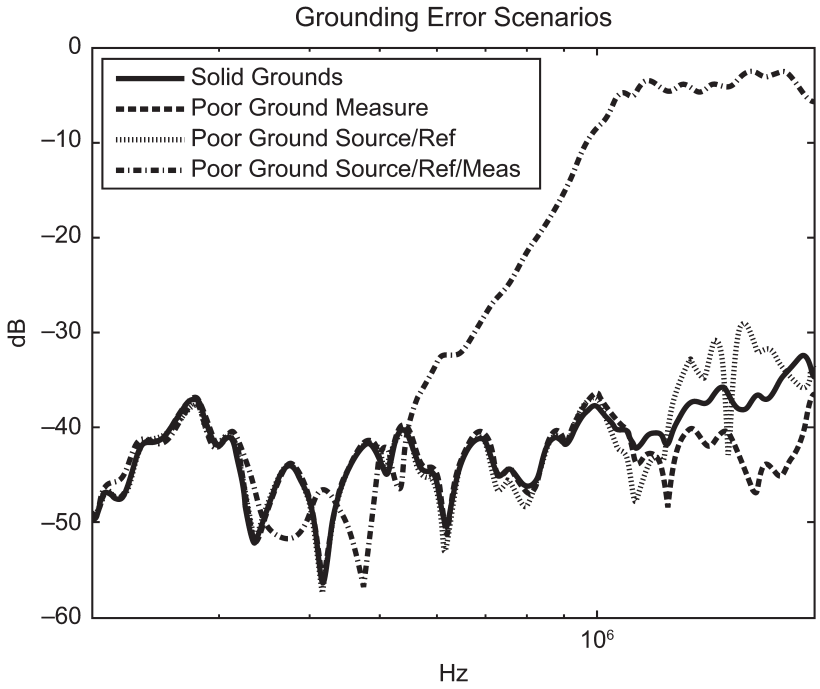


Figure 2—Grounding error scenarios

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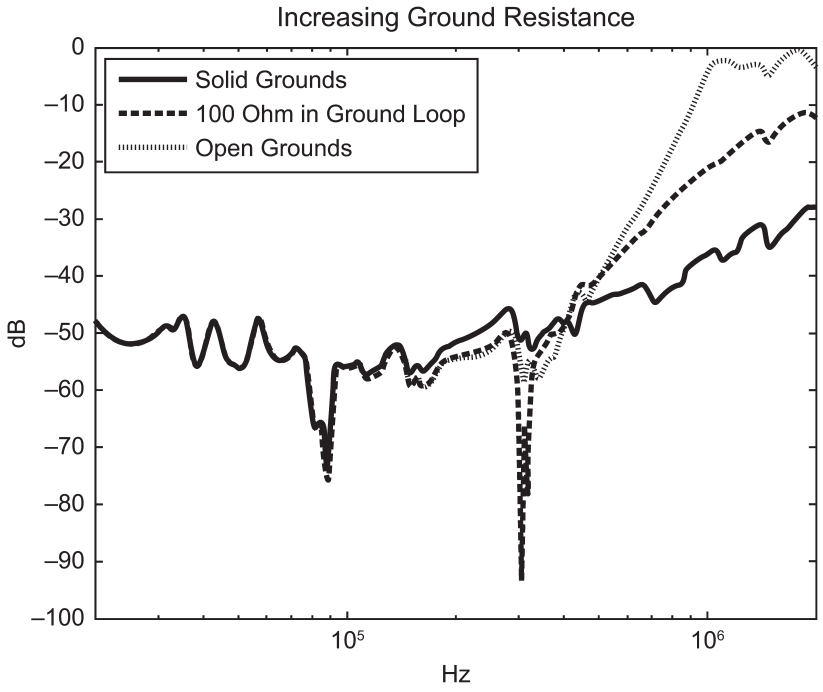


Figure 3—Examples of increasing ground resistance

Poor grounding often affects the measurement at frequencies greater than 200 kHz, and at varying degrees. Figure 2 illustrates the scenario where one, both, or all test leads ground are defective or not applied.

Often, undesired impedance is introduced into the ground loop circuit by rust, paint, oxidation, ground loop length, ground lead placement, or just a poor ground clip connection. The ground loop impedance in Figure 3 was increased by inserting 100 Ω on both test leads between the ground connection on the test leads and the bushing flanges.

When performing a frequency response analysis measurement of a test specimen grounding is extremely critical. For installed transformers the testing may be performed in a noisy environment and the low voltage testing can be influenced. Factory floors are ideal for testing as the noise level is lower, however grounding is still an issue. Every effort should be made to reduce outside influence to acquire the most repeatable result.

The bushings of the transformer can become an antenna allowing power mains frequency interference to distort the results achieved.

Transformers that are removed from service for routine testing should have clean bushing terminals and grounding points. Cleaning of the bushing terminals and grounding points is often necessary.

Testing in a manufacturing environment or newly field assembled transformers may not have a proper ground. It is recommended to help ensure a proper ground.

5.7 Measurement types

5.7.1 Open-circuit measurement

An open-circuit measurement is made from one end of a winding to another with all other terminals floating. The open-circuit test can be applied to both single phase and three phase transformers. Open-circuit tests

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generally fall into the following five winding categories: high voltage, low voltage, tertiary, series, and common. The series and common categories are applied to autotransformers.

Open circuit tests are primarily influenced by the core properties at or around the fundamental power frequency. The open circuit tests can be used in conjunction with exciting current tests in determining failure modes that affect the magnetic circuit of the transformer.

5.7.2 Short-circuit measurement

The short-circuit measurement is made from one end of a high-voltage winding to another while the associated low voltage winding is shorted. For repeatability purposes, it is recommended that all low-voltage windings are shorted on three-phase transformers to create a three-phase equivalent short-circuit model. This helps ensure that all three phases are similarly shorted to give consistent impedance. Any available neutral connections should not be included in the shorting process.

The short-circuit test isolates the winding impedance from the core properties at or around the fundamental power frequency. The short-circuit results should produce similar diagnostic information, as seen in both leakage reactance and DC winding resistance measurements.

5.7.3 Capacitive inter-winding measurement

The capacitive inter-winding measurement also known as the inter-winding measurement is performed between two electrically isolated windings. A capacitive inter-winding measurement is made from one end of a winding and measuring the signal through one of the terminals of another winding, with all other terminals floating. Voltage is applied to one end of a winding, the input, and the response, the output, is measured at another winding, with all other terminals floating. Capacitive inter-winding measurements are capacitive in nature. These measurements exhibit a high impedance at low frequencies (< 100 Hz); the impedance generally decreases as frequency increases. This would include, for example, H1 to X1 on a two-winding transformer or H1 to Y1 on an autotransformer with a tertiary. Note that H1 to X1 on an autotransformer is not an inter-winding measurement but an open-circuit measurement on the series winding.

5.7.4 Inductive inter-winding measurement

The inductive inter-winding measurement, also known as the transfer voltage measurement, is performed between two windings with one end of each winding grounded. All other winding terminals not under test should remain floating. The inductive inter-winding measurement most closely resembles the turns ratio test properties at or around the fundamental power frequency.

5.8 Load tap changer (LTC) and de-energized tap changer (DETC) positions

The tap positions shall be noted on the test report for each test. Generally, tests on windings with a LTC should be in the tap combination that places all sections of the tap windings in the circuit. It is recommended that the LTC be in the extreme raise position.

However, when tests are performed at the LTC neutral tap position, the previous tap position should also be recorded, as the resulting winding connection through the reversing switch will influence the test result. Consistency with previous test applications is key for subsequent analysis. It is recommended that neutral tap position measurements be made after arriving from the raised tap positions.

It is recommended that the DETC be in the position that is dictated by in service conditions.

Transformers in service occasionally have problems due to DETC movement; therefore it is not recommended that the DETC position be altered for the FRA test. The exception is in factory tests on a new transformer, where it can be assumed that the DETC is in satisfactory condition.

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LTC and DETC SUMMARY

— LTC—Extreme raise

— DETC—As found

In a DETC, the process of tap changing is assisted by means of a mechanical rotary or slider switch. The DETC mechanism will often be set just once, at the time of installation and, in very few occasions, it might be changed later to accommodate a long-term change in the system voltage profile.

A DETC that has been sitting on the same position since installation and has not been exercised or maintained mechanically, may have a hard transition due to sluggishness and its repositioning may risk the normal transformer operation.

As stated in 4.3, when the transformer is equiped with a DETC, it is a decision for the transformer owners as to whether they wish to operate the de-energized tap changer. If there is no maintenance record, it is recommended to test on the DETC "as found" position to be able to reflect in the test its "in service" configuration.

5.9 Test connections

The basis of FRA analysis is comparison with reference measurements; if reference measurements are available, it is strongly recommended to repeat those measurements exactly as the reference measurements were set up.

All new measurements should follow the basic principle to measure the windings from "head-to-tail" and consequently the recommended test connections will depend on winding configuration. Table 1 shows the test connections for most phasor group designations (sometimes referred to as "vector groups" as described in IEEE Std C57.12.70).[11]

Table 1—Test connections for phasor group designations

Transformer vector diagram	Phase	HV winding	Phase group/LV winding connection					
			0	2	4	6	8	10
1 ph	A	H1-H2	X1-X2	n/a	n/a	n/a	n/a	n/a
	B	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	C	n/a	n/a	n/a	n/a	n/a	n/a	n/a
YNyn	A	H1-H0	X1-X0	X0-X2	X3-X0	X0-X1	X2-X0	X0-X3
	B	H2-H0	X2-X0	X0-X3	X1-X0	X0-X2	X3-X0	X0-X1
	C	H3-H0	X3-X0	X0-X1	X2-X0	X0-X3	X1-X0	X0-X2
Dd	A	H1-H3	n/a	X1-X2	n/a	n/a	X2-X1	n/a
	B	H2-H1	n/a	X2-X3	n/a	n/a	X3-X2	n/a
	C	H3-H2	n/a	X3-X1	n/a	n/a	X1-X2	n/a

Table continues

[11]Information on references can be found in Clause 2.

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Table 1—Test connections for phasor group designations *(continued)*

Transformer vector diagram	Phase	HV winding	Phase group/LV winding connection					
			0	2	4	6	8	10
Dd	A	H1-H2	X1-X2	n/a	X3-X1	X2-X1	n/a	X1-X3
	B	H2-H3	X2-X3	n/a	X1-X2	X3-X2	n/a	X2-X1
	C	H3-H1	X3-X1	n/a	X2-X3	X1-X3	n/a	X3-X2

Transformer vector diagram	Phase	HV winding	Phase group/LV winding connection					
			1	3	5	7	9	11
YNd	A	H1-H0	X1-X2	X3-X2	X3-X1	X2-X1	X2-X3	X1-X3
	B	H2-H0	X2-X3	X1-X3	X1-X2	X3-X2	X3-X1	X2-X1
	C	H3-H0	X3-X1	X2-X1	X2-X3	X1-X3	X1-X2	X3-X2
Dy	A	H1-H3	X1-X3	n/a	n/a	X3-X1	n/a	n/a
	B	H2-H1	X2-X1	n/a	n/a	X1-X2	n/a	n/a
	C	H3-H2	X3-X2	n/a	n/a	X2-X3	n/a	n/a
Dy	A	H1-H2	n/a	n/a	X3-X1	n/a	n/a	X1-X3
	B	H2-H3	n/a	n/a	X1-X2	n/a	n/a	X2-X1
	C	H3-H1	n/a	n/a	X2-X3	n/a	n/a	X3-X2
Dyn	A	H1-H3	X1-X0	n/a	n/a	X0-X1	n/a	n/a
	B	H2-H1	X2-X0	n/a	n/a	X0-X2	n/a	n/a
	C	H3-H2	X3-X0	n/a	n/a	X0-X3	n/a	n/a
Dyn	A	H1-H2	n/a	n/a	X0-X1	n/a	n/a	X1-X0
	B	H2-H3	n/a	n/a	X0-X2	n/a	n/a	X2-X0
	C	H3-H1	n/a	n/a	X0-X3	n/a	n/a	X3-X0

Note 1—Transformers with tertiary windings are tested as a two-winding transformer, looking at the configuration between HV and tertiary windings. It is recommended that tertiary delta windings are closed and ungrounded (if bushings are available externally).
Note 2—The highlighted area corresponds to non-applicable connections.

Example testing a 2-winding Dyn5 transformer.

a) For open-circuit test mode:

   1) Look at transformer vector diagram column—Last row in that column shows Dyn.

   2) Look at the secondary winding phase deviation columns. In this example the number is 5.

   3) Follow the instruction for HV side first connection on phase A: Generator and reference leads to first terminal H1, measurement lead to H2.

   4) Complete HV measurements for phases B and C following connections as presented in the table.

   5) Follow the instructions for LV side corresponding phase A connections: Generator and reference leads to first terminal X0, measurement lead to X1.

   6) Complete LV measurements for phases B and C following connections as presented in the table.

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b) For short circuit test

1) Look at transformer vector diagram column—Last row in that column shows Dyn.

2) Look at the secondary winding phase deviation columns. In this example the number is 5.

3) For all tests: Short X1-X2 and X3. Do not include X0.

4) Follow the instruction for HV side first connection on phase A: Generator and reference leads to first terminal H1, measurement lead to H2.

5) Complete HV measurements for phases B and C following connections as presented in the table.

If transformer-winding configurations are required (which are not covered in Table 1), please refer to the transformer nameplate. The configuration vectors should determine the test procedure.

The test connections described here do not include repeat tests for different tap positions. Bushings not under test, including neutrals, should be disconnected from the ground unless grounding is required under local recommendations and/or guidelines.

At a minimum, it is recommended that all open-circuit tests and all short-circuit tests be performed as shown in Table 2, Table 3, and Table 4, which are the tests highlighted and within the bold border should be performed.

Alternatives to the test sequences suggested below can be selected based on the recommendations of the transformer manufacturer, the test equipment manufacturer, the test equipment user, and the type of test voltage applied.

The most common two winding and autotransformers configurations are shown below in the test connection tables. The tables are as follows:

— Table 2—Two winding transformers

— Table 3—Autotransformer without tertiary

— Table 4—Autotransformer with tertiary

Table 2—Two winding transformers—15 tests

Test type	Test #	Dyn1	YNd1	Dd0	Yy0	1ϕ
HV open circuit (OC)	1	H1-H3	H1-H0	H1-H2	H1-H0	H1-H2
All other terminals	2	H2-H1	H2-H0	H2-H3	H2-H0	[H1-H0]
Floating	3	H3-H2	H3-H0	H3-H1	H3-H0	
LV open circuit (OC)	4	X1-X0	X1-X2	X1-X2	X1-X0	X1-X2
All other terminals	5	X2-X0	X2-X3	X2-X3	X2-X0	[X1-X0]
Floating	6	X3-X0	X3-X1	X3-X1	X3-X0	
Short circuit (SC)	7	H1-H3	H1-H0	H1-H2	H1-H0	H1-H2
Short	8	H2-H1	H2-H0	H2-H3	H2-H0	Short
[X1-X2-X3][a]	9	H3-H2	H3-H0	H3-H1	H3-H0	[X1-X2][a]
Capacitive inter-winding	10	H1-X1	H1-X1	H1-X1	H1-X1	H1-X1
All other terminals	11	H2-X2	H2-X2	H2-X2	H2-X2	
Floating	12	H3-X3	H3-X3	H3-X3	H3-X3	

Table continues

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Table 2—Two winding transformers—15 tests *(continued)*

Test type	Test #	Dyn1	YNd1	Dd0	Yy0	1ϕ
Inductive inter-winding High (H) to low (L) Ground (H- and X-)[b]	13	H1-X1	H1-X1	H1-X1	H1-X1	H1-X1
	14	H2-X2	H2-X2	H2-X2	H2-X2	Ground
	15	H3-X3	H3-X3	H3-X3	H3-X3	[H2, X2]

[a]Indicates short circuit tests terminals are shorted together, but not grounded. The neutral is not included for 3ϕ Wye connections, but may be included for 1ϕ connections.
[b]Denotes other end of winding; opposite of the reference and measure connections.

Table 3—Autotransformer w/o tertiary—12 tests

Test type	Test #	3ϕ	1ϕ
Series winding (OC)	1	H1-X1	H1-X1
All other terminals	2	H2-X2	
Floating	3	H3-X3	
Common winding (OC)	4	X1-H0X0	X1-H0X0
All other terminals	5	X2-H0X0	
Floating	6	X3-H0X0	
Short circuit (SC)	7	H1-H0X0	H1-H0X0
High (H) to low (L)	8	H2-H0X0	Short
Short [X1-X2-X3][a]	9	H3-H0X0	[X1-H0X0][a]
Inductive inter-winding	10	H1-X1	H1-X1
High (H) to low (L)	11	H2-X2	Ground
Ground (H0X0)	12	H3-X3	[H0X0]

[a]Indicates short circuit tests: terminals are shorted together, but not grounded. The neutral is not included for 3ϕ Wye connections, but may be included for 1ϕ test connections.

Table 4——Autotransformer with tertiary – 33 tests

Test type	Test #	3ϕ	1ϕ
Series winding (OC)	1	H1-X1	H1-X1
All other terminals	2	H2-X2	
Floating	3	H3-X3	
Common winding (OC)	4	X1-H0X0	X1-H0X0
All other terminals	5	X2-H0X0	
Floating	6	X3-H0X0	
Tertiary winding (OC)	7	Y1-Y2	Y1-Y2
All other terminals	8	Y2-Y3	(Y1-Y0)
Floating	9	Y3-Y1	
Short circuit (SC)	10	H1-H0X0	H1-H0X0
High (H) to low (L)	11	H2-H0X0	Short
Short [X1-X2-X3][a]	12	H3-H0X0	[X1-H0X0][a]
Short circuit (SC)	13	H1-H0X0	H1-H0X0
High (H) to tertiary (Y)	14	H2-H0X0	Short
Short [Y1-Y2-Y3][a]	15	H3-H0X0	[Y1-Y2][a]
Short circuit (SC)	16	X1-H0X0	X1-H0X0
Low (L) to tertiary (Y)	17	X2-H0X0	Short
Short [Y1-Y2-Y3][a]	18	X3-H0X0	[Y1-Y2][a]

Table continues

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Table 4—Autotransformer with tertiary – 33 tests *(continued)*

Test type	Test #	3ϕ	1ϕ
Capacitive inter-winding	19	H1-Y1	H1-Y1
High (H) to tertiary (Y)	20	H2-Y2	
All terminals float	21	H3-Y3	
Capacitive inter-winding	22	X1-Y1	X1-Y1
Low (L) to tertiary (Y)	23	X2-Y2	
All terminals float	24	X3-Y3	
Inductive inter-winding	25	H1-X1	H1-X1
High (H) to low (L)	26	H2-X2	Ground
Ground (H0X0)	27	H3-X3	[H0X0]
Inductive inter-winding	28	H1-Y1	H1-Y1
High (H) to tertiary (Y)	29	H2-Y2	Ground
Ground (H0X0 and Y-)[b]	30	H3-Y3	[H0X0, Y2]
Inductive inter-winding	31	X1-Y1	X1-Y1
Low (L) to tertiary (Y)	32	X2-Y2	Ground
Ground (H0X0 and Y-)[b]	33	X3-Y3	[H0X0, Y2]

[a]Indicates short circuit tests: terminals are shorted together, but not grounded. The neutral is not included for 3ϕ Wye connections, but may be included for 1ϕ test connections.
[b]Denotes other end of winding; opposite of the reference and measure connections.

5.9.1 FRA shipping configuration

Performing the test under the same conditions is important for comparison purposes since the premise of the test is to detect changes in the winding geometries. For this reason, the optimal method for performance of the FRA test to verify that no changes have taken place during transportation is to require the test be performed at the factory on the transformer in its shipping configuration. This will then allow the FRA test to be performed at any time during transit, and provide data for comparison to potentially aide in the determination of when shipping damage may have occurred.

It is important that the “shipping configuration” of the equipment be identified and duplicated exactly for pre and post movement tests. Four primary items that should be specified for shipping configuration tests are as follows:

a) Oil level and presence
b) De-energized tap changer tap position
c) Load tap changer tap position
d) Bushings and placement
e) Test/transformer and earthing configuration

Additionally, it may be necessary to identify more than one shipping configuration during transport of the transformer. The transformer may undergo several legs of transport to include truck, ship, railroad, crane off-loading, and final movement to the installation pad. Any of these transportation legs can exert undue physical shock to the transformer. It is important to identify these different legs of transport and identify the entities who have legal responsibility. Testing before and after transport legs that have different legal custodies is an additional option that can be considered. Since the transformer may be in different shipping configuration due to physical constraints, shipping configurations should be clearly identified for each transportation leg.

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For new equipment, this may require the performance of two FRA tests after receipt of the equipment at the final destination; 1) one test with the transformer in its shipping configuration, 2) and one test with the transformer assembled and oil-filled as required for insulation resistance testing, to be used as baseline data for future testing. If no shipping damage is suspected, the test in the as shipped configuration may not be necessary as a receipt test.

5.9.2 Oil level and presence

The equipment may be shipped with or without oil depending on size and shipping restrictions. It may also be filled then drained prior to shipment, thus leaving residual oil within cellulosic insulation in the windings. Oil is known to cause variations in the FRA test readings across a broad range of test frequencies. The reason for the variation is that the dielectric constant of oil is different than that of air and changes the capacitance of any insulation system within equipment involving oil. There is also a difference in dielectric constants between types of fluids that should be considered. The FRA transfer function measured is a complex network measurement that not only involves bulk measurements but also turn-to-turn. Therefore, all the unique resonances formed by this network will be shifted higher in frequency by the absence of oil. There are also indications that some equipment will display unique FRA characteristics with and without oil.

If the equipment is to arrive drained of oil, the shipping configuration should specify that it should be tested pre and post movement without oil. If the equipment is to be shipped after being drained of oil, it should be tested pre-movement without oil. Testing the unit prior to shipment in this case without oil and prior to a first fill, may not be adequate and could lead to false failures due to residual oil being held in the windings, or additional oil draining from the winding during weeks of shipment. If the equipment is to be shipped with oil, it should be fully filled for both pre and post movement tests. If the equipment is to be shipped partially filled, it should be tested with the same level of oil, or preferentially after oil has been added. Confirming that oil is at the same level before and after transportation for partially filled transformers can be difficult and sometimes leads to incorrect assessments.

5.9.3 De-energized and on-load tap changer position

Both the DETC and LTC position will change the FRA test results as the number of winding turns and tap changer components are different for different positions. As a result, the equipment should be tested in the exact same positions for every pre and post movement leg. It is recommended to test the equipment in the nominal DETC tap position and the LTC fully raised position.

5.9.4 Bushings and placement

For performance of the FRA test with the equipment in its shipping configuration and with bushings removed for shipment, it may be beneficial to install a small FRA test bushing in each removed bushing's cover to plate to facilitate testing. The FRA test bushings should be of robust construction so as to minimize the potential of damage during shipment, and should be gasketed to maintain the shipping pressure on the tank. Typical ratings for these bushings would be 1.2 kV voltage class, 30 kV BIL. The winding leads should be connected to the FRA test bushings, and sufficiently braced to avoid movement during transit. Any jumpers used for connection of the winding leads to the test bushings should be appropriately marked, to assure removal during the final equipment assembly after transportation. If the equipment is to be tested with the as-installed bushing, the bushings should be placed in the same tank positions. The different capacitance-ground values of condenser style bushings can sometimes influence the FRA test results.

6. Test documentation

6.1 Introduction

The frequency response measurement is one that provides a wealth of information regarding the internal geometry of a transformer.

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To gain value from the measurement, it is important that measurements are made in a consistent manner. Unlike power factor measurements, where it is possible to make consistent measurements based on an analysis of the transformer design and some basic electrical engineering "first principles," FRA requires measurements to be made consistent with previous measurements or with those on similar units.

Accordingly, it is important that any test parameter that could affect test results is recorded in a clear, concise, format such that the FRA signatures can be reconstructed without the use of proprietary software.

6.2 Test records

A number of different elements of a particular test need to be recorded – this helps ensure that the same test setup is possible on subsequent tests, and with respect to tests on other transformers.

The details given here are separated out into separate elements of a test set up, but should not be taken as a prescription of how data should be stored.

In each area, the suggested minimum information required is marked as "required;" some other data can be inferred from asset management systems within the transformer owner's organization.

6.2.1 Digital data format usage and storage

Data should be stored in digital format and viewable in most standard applications to allow flexible analysis of the results, comparison between results from different test systems, and inclusion in reports. Because FRA records can be stored for many years before being recalled for comparison, it is recommended to avoid any file format associated with current applications or instruments, as these can change with time or might not be supported 15 or 20 years after they have been stored. Because of this, we suggest that the XML file format is used. It is a well-known and open format which has been available since the 1990s.

6.2.2 Data elements

The object of a test is a transformer. Clearly, nameplate data for the transformer should be recorded, but there are other factors that may vary between tests, which may affect results. These include the following:

a) **Transformer**: What is tested, nameplate data

b) **Test equipment**: Especially if we are comparing data from a number of different OEM sets

c) **Test organization**: To detect any systematic errors, which may result from poor procedures

d) **Test setup**: State of transformer, oil level, bushings, tap position, temperature, etc.

e) **Test results**: Date, time, frequency, magnitude and phase angle

6.2.3 Transformer data

The "static" data which does not vary between tests on a particular transformer includes the following:

a) Manufacturer—Required

b) Year of manufacture

c) Serial number—Required

d) MVA rating for various cooling modes—ONAN, OFAF, ODAF, etc.

e) Voltage rating—HV, LV1, LV2, tertiary

f) Special ID (transformer ID)—Required

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g) Vector group configuration

h) Number of phases

i) Impedance—HV-LV1, HV-LV2, HV-tertiary, LV-tertiary

j) Transformer type—GSU, power XFMR, dist, furnace, etc.

k) Winding type—Auto, double wound, etc.

l) Transformer construction—Core form, shell form, winding type, etc.

m) LTC fields—Manufacturer, serial number, manufacture year, range, notes

n) DETC fields—Manufacturer, serial number, manufacture year, range, notes

o) Free-form data entry fields for user-specific details

p) Connection and grounding condition of any buried tertiary windings

6.2.4 Test equipment data

The following test data equipment records are key to identifying the equipment used for the measurement:

a) Equipment name and model number

b) Manufacturer

c) Test equipment serial number

d) Calibration date

e) Measurement impedance

f) Free-form data entry fields for user-specific details

6.2.5 Test organization data

The following records identify specific information related to the organization and personnel involved in the testing:

a) Company—Required

b) Location—Required

c) Operator

d) Free-form data entry fields for user-specific details

6.2.6 Test set-up data

The following elements may vary between tests on a given transformer:

a) Oil temperature

b) Oil level

c) Oil status (whether immersed or not)—Required

d) External circuits connected-bushings (length of bus, etc.)—Required

e) Reason for test

f) Free-form data entry fields for user-specific details

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6.2.7 Test results

The test results contain the actual data points and test setup data specific for the measurement. Every measurement of the following in the test is represented by a data block:

a) Name

b) Date and time of measurement—Required

c) LTC position, including previous position if at neutral—Required

d) DETC position – Required

e) Measurement type (OC, SC, capacitive inter-winding, inductive inter-winding)—Required

f) Terminals (e.g., source, reference, measure, shorted, grounded)—Required

g) Data points: Frequency, amplitude, phase angle

h) Free-form data entry fields for user-specific details

i) Applied test voltage—Required

7. Measurement analysis and interpretation

7.1 Introduction

This clause looks at the analysis value from FRA measurements. Two basic analysis strategies are deployed in this clause: identifying expected trace characteristics, and comparing a trace with reference results. The primary goal of FRA analysis is to determine the physical condition of the transformer, thus confirming internal components have not moved as a result of transportation, severe insulation damage, or fault currents.

7.2 Trace characteristics

Since transformer designs and applications vary, the FRA results exhibit diverse properties and characteristics. However, the FRA trace, over specified frequency ranges, has a degree of predictability; caused by low frequency core effects, main winding effects, and short circuit responses. These expected responses can be used to identify basic problems that may exist within a transformer. In addition, different winding configurations generally exhibit distinct patterns due to their relationship between phases and the core. With an understanding of the relationship between frequency range and transformer configuration, physical deformations can be narrowed down to specific sections of the transformer.

The different measurement types produce different characteristic trace wave shapes, but the expected changes in the traces are generally similar. Some trace characteristics indicate more significant winding deformation and insulation degradation than others. One of the most significant change indicators is the presence of an additional resonance peak(s) in the magnitude trace or the loss of an existing resonance peak(s). The next most significant change indicator is usually a shift in frequency for an existing resonance peak(s). These are usually examples of winding deformation. Another trace characteristic indicator is when a trace continues to increase in magnitude compared with the original trace, while maintaining a similar shape, as frequency increases. This may be an indicator of winding looseness, especially for higher frequencies. Most significant trace magnitude differences are also accompanied by significant trace phase angle differences.

The type of FRA test greatly affects the expected trace characteristics. Specific characteristic can be identified depending on the type of test.

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7.2.1 Open-circuit test

The open-circuit test is the most common FRA test performed. Similar to an excitation current test, each winding is excited individually while all other terminals float. Voltages with reference to ground are measured at each end of the excited winding. This test is most influenced by the effects of the core, main windings, tap winding, and tap leads. Depending on the winding configuration, the low frequency section (core) will usually take on a distinct shape.

High-voltage windings display the most distinguishable pattern in the low frequency region. Wye connected windings often show a single distinct resonance or two closely placed null resonance between 20 Hz and 5 kHz. Delta connected windings will usually show two further resonances spaced apart in this same frequency range. Low or tertiary voltage windings tend to follow the same pattern, but can be heavily influenced by the high voltage winding.

The mid frequency region is dominated by the network impedances created by the winding stack itself. There could be either a complex series or simple series of resonances created.

Above 500 kHz, the results will continue-be influenced by the main winding's impedance network, but tends to be more sensitive to smaller geometric sections such as tap windings, tap leads, and displaced turns.

7.2.2 Short circuit test

Short-circuit tests are designed to allow for the inspection of the winding without the influence of the core. The transformer core's reluctance circuit influences the lowest frequencies. To analyze the bulk inductance of the winding without the influence of the core, the secondary windings are short circuited. This configuration is similar to a short circuit impedance test or leakage reactance test.

The short-circuit test's trace characteristic is similar to the open circuit test with the exception of the low frequency region. At these low frequencies, the increased impedance associated with the core reluctance is removed. This results in an overall increase in the FRA trace's magnitude in this region as well a more inductive phase angle. The shape in the low frequency is dominated by the first order roll-off characteristic of the windings' main inductance. If the results are examined on a Bode plot, a predictable inductive roll-off section can often be identified as is expected with first-order systems.

7.2.3 Capacitive inter-winding

Capacitive inter-winding tests measure the network impedance between two voltage class windings. Given that this measurement does not offer a galvanic connection between the windings involved in the measurements, as the two windings are isolated, this trace shows an increasing magnitude as a function of frequency. At the lowest frequencies, the measurement is highly capacitive as would be expected by the open circuit configuration between the two test terminals. As the frequency is increased, a series of resonances will be formed with an overall trend of increasing magnitude. These resonances are formed by the network impedances between the two voltage class windings.

7.2.4 Inductive inter-winding

The inductive inter-winding test is designed to measure the voltage ratio between two windings. In this test configuration, the two vector matching terminals are measured across while grounding the associated leg terminal on each side. This configuration is similar to a single phase turns ratio test. The trace is dominated by a flat magnitude response at the lowest frequencies. The magnitude in the low frequency region is equal to the voltage ratio formed by the voltage class ratio between the two windings under test. At the higher frequencies, a series of resonances are formed that are generally not examined, as it is not the primary focus of this test.

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7.3 Trace comparison

Trace comparison is the primary method for the analysis of FRA results. Comparisons can be made against baselines and previous data, sister unit results, or between phases. Traces can also be examined for expected patterns as mentioned in 7.2; this type of comparison requires more experience but can prevent erroneous data from being collected. Assuming the test equipment is compatible, connections are the same, and there is not residual magnetism, the initial expectation is that any data comparison should result in near perfect overlays.

Various tools and algorithms are available for comparing and analyzing FRA measurements.

7.3.1 Plot inspection

Plot inspection involves plotting the FRA transfer function and examining the results. This method of analysis is the most common and relies upon a reasonable level of skill. Specialized computer software or spreadsheets are commonly used to present data. Common practice is to display the plot in decibels (dB) versus logarithmic frequency. This type of plot is also known as a Bode plot. Bode plots allow easier inspection of resonances as it simplifies the display of exponentially changing impedances and are used cross-industry for the analysis of transfer functions. Plots can also be graphed as impedance or admittance versus frequency; either linear or logarithmic scales can be applied.

The plots are inspected for expected general patterns due to transformer configuration and comparing the results with other traces. The other traces can be different phases, sister units or baseline results. This method relies upon careful scrutiny of the trace resonances and magnitude/phase angle deviations. Please refer to 7.2 for a discussion on frequency range-component correlation and expected patterns. Deviations noted due to comparison with another unit or phase could indicate physical deformation of a transformer's component.

7.3.2 Difference plotting

Difference plotting was one of the first methods of analysis used for transformer FRA tests. This type of analysis takes two traces for comparison and subtracts one from the other. A difference trace is the result. Regions in the difference plot that deviate from zero (i.e., no variation between the two traces) could indicate problems and would require greater scrutiny. Resonance shifts or deviation in FRA magnitude would show the greatest variation on the plot. This method does not necessarily assist ascertaining the cause of the problem, but can assist in identifying suspect frequency regions. Difference plotting is not used on phase angle plots due to the sensitivity of the phase angle measurement.

7.3.3 Correlation coefficients

Correlation coefficients provide an indication of similarity between two traces. Another common term for correlation is the Pearson's correlation coefficient or Pearson's product moment correlation. Correlation allows for the expression of how random (or similar) a range of numbers is. Correlation has a value between −1 and 1. If the value is zero (0), this indicates complete randomness between the two traces under inspection. The closer the correlation is to one (1), the more similar the two traces are. If the value is −1.0 this indicates a complete inverse relationship. The goal for analysis is to get a number as close to one (1) as possible, indicating correlation between the two traces.

Individual frequency ranges can be analyzed for correlation by changing the frequency bounds of the correlation algorithm. Correlation is an algorithm, and is often calculated with the help of specialized FRA software or commercially available mathematical programs. Correlation coefficients are calculated by determining the covariance of the two traces under test. Covariance is expressed as σ_{xy}. Covariance calculates how much two traces deviate from each other. This is expressed in Equation (1) and Equation (2) below. In the case of isolating certain regions, the bounds would be changed from plus or minus infinity to the spectrum under question.

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$$\mathrm{cov}(X, Y) = \sigma_{xy} \tag{1}$$

$$\sigma_{xy} = E[(X - \bar{X})(Y - \bar{Y})] \tag{2}$$

$$\rho = \frac{\sigma_{xy}}{\sigma_x \sigma_y} \quad -1 \leq \rho \leq 1 \tag{3}$$

where

σ_{xy} is the covariance of the x and y measured values
E[ ] is the mean of the covariance of the X and Y variables
X is trace X and its measured values
$\bar{X}$ is the mean of the X trace measured values
Y is trace Y and its measured values
$\bar{Y}$ is the mean of the Y trace measured values
ρ is the correlation constant which is greater than or equal to −1 and less than or equal to 1
σ_x is the standard deviation of trace x measured values
σ_y is the standard deviation of trace y measured values

Finally, the correlation constant ρ is calculated by taking the covariance of the two traces under question and then dividing them by the product of the standard deviation of each trace, thus normalizing the constant for easy inspection.

7.3.4 Baseline data

An FRA baseline measurement can be produced in the factory when the transformer has been filled with oil and dressed as part of factory commissioning tests, or as part of routine testing for units already in-service, or at an earlier date in the substation.

Baseline or previous data should be repeatable. If internal movement or change does not occur within the test specimen, the matched traces should overlay well. Matched traces are defined as FRA results obtained from the same test terminals and transformer configurations. An example would be two scans collected from the same winding, such as H1-H3, on different test dates.

Data that is collected before and after transformer relocation is expected to overlay well. Any variance in such comparisons indicates a problem. One exception, caused by the magnetic circuit, occurs at low frequencies and should be considered during evaluation. Magnetization can cause the lower frequencies of the trace to be slightly offset in certain cases. If residual magnetism is known to be present, less scrutiny should be placed on this comparison. Large temperature difference, typical much more than 10C, between two measurements will slightly influence the response at higher frequencies.

As stated in 5.3, it should be noted that the LTC and DETC position influences the results. If the test results are obtained in different tap positions, expect variation.

7.3.5 Similar units

Similar unit results are expected to compare well. Genuinely similar units show very little difference between matched scans. Genuinely similar units are considered to be identical in manufacturer, design, and construction. Care should be taken to assess if the similarity of a unit is really genuine. Even if a transformer appears identical in all aspects, it is possible that the two units may be constructed with some differences.

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This is most prevalent when a utility purchases two identical transformers several months or years apart. The manufacturer may have made modifications to the design that does not change the functionality of the transformer, but could lead to variations in the traces.

All tests on similar units should be conducted with the LTC and DETC in the same position. The low frequency and mid frequency regions governed by the core's magnetic circuit and main windings should have the closest similarity. There may be variations in the higher frequencies due to subtle changes in tap lead placement and routing. The more sister units that can be compared, a better feel for the overall stability of their traces can be ascertained. As such, it is beneficial to plot several sister units on the same graph for inspection, such as three single-phase transformer installations.

7.3.6 Phase

Phase comparisons are the most difficult and require the greatest amount of experience to conduct properly. In addition, some transformer designs do not have completely symmetrical designs. In these cases, there will always be some phase-to-phase variations. Because of this, it is paramount that benchmark data be obtained as early in a transformer's life as possible. In most cases, different phases will overlay with reasonable similarity

The center phase, especially in core type transformers, exhibits the most deviation when comparing all three phases. Often, the two outer phases compare well. Different flux paths seen by each phase contribute to the observed differences. The effects of the core are expected at the lower frequencies; however, the core influence can overlap into the higher frequency range.

The actual windings of a three-phase transformer can be almost identical, but the connection scheme between phases can be different. As an example, the phases of a wye winding are all at different distances from the neutral; LTC connections fall into the same category. Thus, since the windings are not equilaterally spaced, the varying lead lengths entering and leaving the windings influence the individual transfer function of each winding. Overall phase symmetry appears to be a function of the overall physical size and complexity. An example that requires consideration is when a tertiary winding is present, that has only one corner grounded and is not accessible.

When no baseline test exists, and there is no genuine sister unit available to compare against, it is still possible to determine obvious failure modes by cross phase comparison of the same voltage windings.

7.4 FRA relationship to other transformer diagnostics

The FRA results (depending on the particular test connections) can be used to confirm the results of other diagnostic tests. These tests include:

— Single phase exciting current

— Turns ratio

— Short circuit impedance (leakage reactance)

— DC winding resistance

7.4.1 Single-phase exciting current

Single-phase exciting current test results can be compared with the FRA's low frequency region for the open circuit test that is applied to the HV winding. The open circuit test is heavily influenced by the core properties at or around the fundamental power frequency. Even though the core properties are dependent on the applied voltage level, the FRA results are expected to produce a pattern similar to the single-phase exciting currents test. When examining the magnitude plots in the low frequency region, their decibel, or ohm levels should mimic the pattern exhibited by the same connection excitation current test at the excitation frequency.

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7.4.2 Turns ratio

The inductive inter-winding test most closely resembles the turns to ratio test properties at or around the fundamental power frequency. The transfer function unit, dB or Ω can be interpreted to match the turns ratio results. Several frequencies at or around the fundamental power frequency should be averaged to estimate the turns ratio value.

7.4.3 Short-circuit Impedance (leakage reactance)

The short-circuit test produces a response at lower frequencies that is associated with the leakage channel of the windings. The phase pattern of these results can be compared to individual 3 phase equivalent results, assuming they were performed on the same tap positions. Any differences between phases in this frequency range should be checked with a leakage reactance test.

7.4.4 DC winding resistance

If the short-circuit test produces a horizontal response at frequencies, less than 30 Hz, then the FRA results can be compared to the DC winding resistance results. Any differences between phases at these low frequencies should be checked with a DC winding resistance test.

7.5 Failure modes

In general, the FRA test is sensitive to defects that cause geometric change(s) within a transformer. Any defect of this kind is referred to as failure mode even though such defect does not necessarily lead to a catastrophic failure of equipment. In fact, the popularity of the FRA test has been driven by the desire to detect mechanical failures within a transformer. Failure modes are not exclusive to geometric variations within a transformer and can include variation in the core's magnetic circuit and contact resistance.

FRA test variations can be caused by a single type of failure or a combination of two or more. Failure due to faults creates high over-currents through the transformer. As a result, the transformer experiences strong and often violent electromagnetic forces. These violent events can often lead to compounded failure modes. These compounded events can complicate the FRA analysis but often helps to better understand the condition of the transformer. The following are listed known failure modes with an explanation of each:

— Radial "hoop buckling" deformation of winding
— Axial winding elongation "telescoping"
— Overall—Bulk and localized movement
— Core defects
— Winding turn-to-turn short circuit
— Residual magnetization

It should be noted that the effects of deformations on the FRA measurements vary with transformer type and design. The same deformation type may affect different transformers differently. Frequency ranges for failure modes given in tables below are approximate and might be some overlap between ranges.

— Radial "hoop buckling" deformation of winding
— Axial winding elongation "telescoping"
— Overall—bulk and localized movement

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7.5.1 Radial winding deformation—"Hoop buckling"

Radial winding deformation or "hoop buckling" is a winding compressive failure that is characterized by a pronounced change to the windings radial geometry. This type of failure can result from the high current electromagnetic forces caused by high over-current faults. The winding is subjected to high radial compressive (inwards) forces and will end up "buckling" along its entire length. The forces are concentrated on the inner windings. Radial winding deformation occurs in two forms, free and forced.

Radial deformation can occur on any winding, however the most common radial failure occurs on the inner LV winding of core form units.

Radial winding deformation affects the FRA measurements as shown in Table 5:

Table 5—Radial winding deformation

Frequency range	Radial winding deformation (assuming no other failure modes exist)
20 Hz to 10 kHz	**Open circuit tests:** This region (core region) is generally unaffected during radial winding deformation. **Short circuit tests:** Results in an increase in impedance. The FRA trace for the affected phase generally exhibits slight attenuation within the inductive roll to off portion of the affected winding. This is a key focal point for the analysis of radial deformation. All measurements are expected to be within +/− 0.2 dB. Measurements not meeting this criterion should be investigated.
5 kHz to 100 kHz	**Open circuit and short circuit tests:** The bulk winding range can shift or produce new resonance peaks and valleys depending of the severity of the deformation. However, this change is minimal and difficult to identify. The changes will be greater for the affected winding, but it is still possible to have the effects transferred to the other winding(s). The response in the bulk region should be used as secondary evidence to support the analysis.
50 kHz to 1 MHz	**Open circuit and short circuit tests:** Radial winding deformation is most obvious in this range. All radial deformation cases exhibit some noticeable shift in this range. It can shift or produce new resonance peaks and valleys depending on the severity of the deformation. The changes will be greater for the affected winding, but it is still possible to have the effects transferred to the other winding(s).
> 1 MHz	**Open circuit and short circuit tests:** This range is generally unaffected. Test leads grounding, and other smaller components generally affect this range. However, severe deformation can extend into this range. This range can be used to validate a good measurement with respect to radial deformation. If the other ranges indicate radial deformation, and this high frequency range compares well, then user error can be dismissed, and radial deformation should be suspected.

Typical "radial" winding deformation results are shown in Figure 4 and Figure 5.

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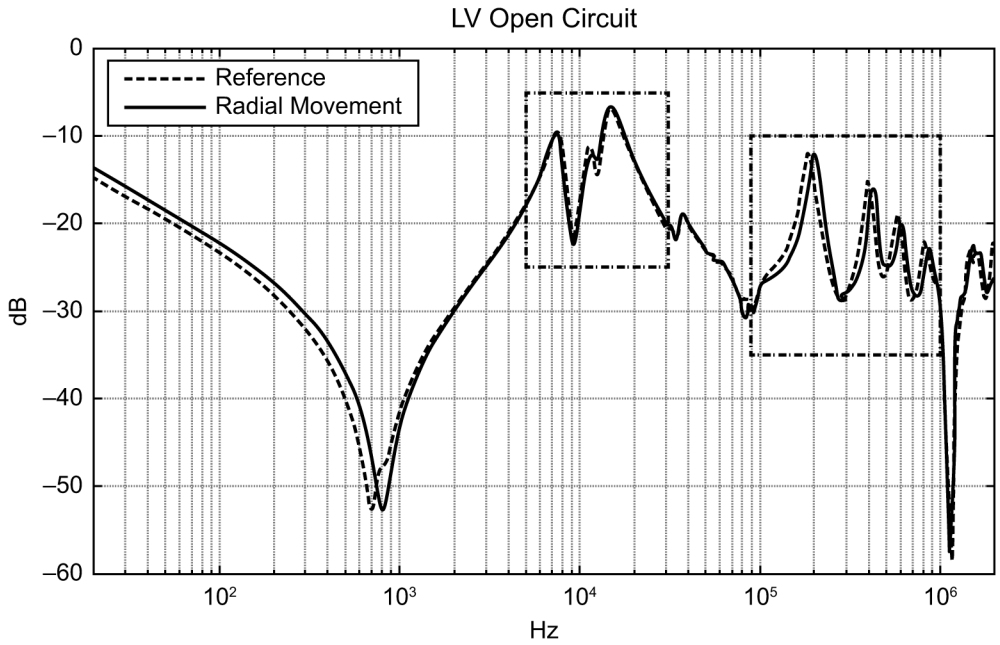


Figure 4—Radial movement response from LV open-circuit test

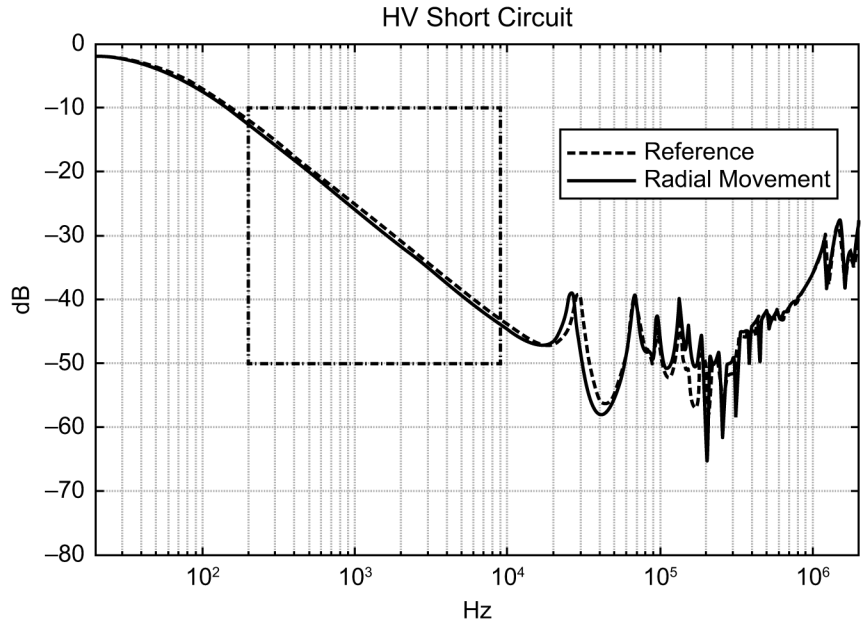


Figure 5—Radial movement response from HV short-circuit tests

7.5.1.1 Case study 1—Radial deformation

Figure 6 and Figure 7 were collected from a transformer that exhibited radial deformation on the LV winding. Phase B was most affected. A change/shift in the 50 kHz to 1 MHz frequency range in Figure 6 is obvious along with the inductive roll-off differences in the lower frequency range in Figure 7. It is typical that the affected phase exhibits an increased impedance during the inductive roll-off.

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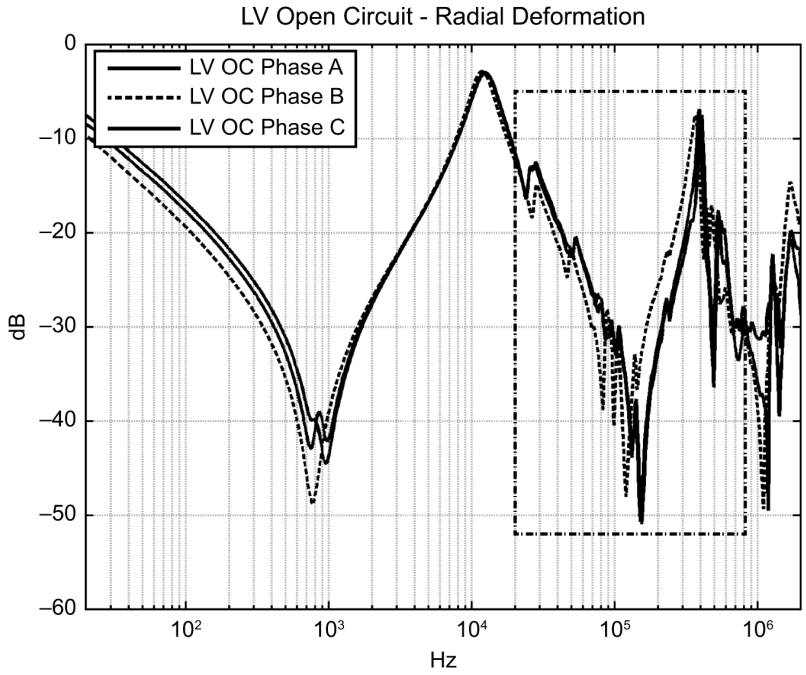


Figure 6—Radial movement response from HV open-circuit tests

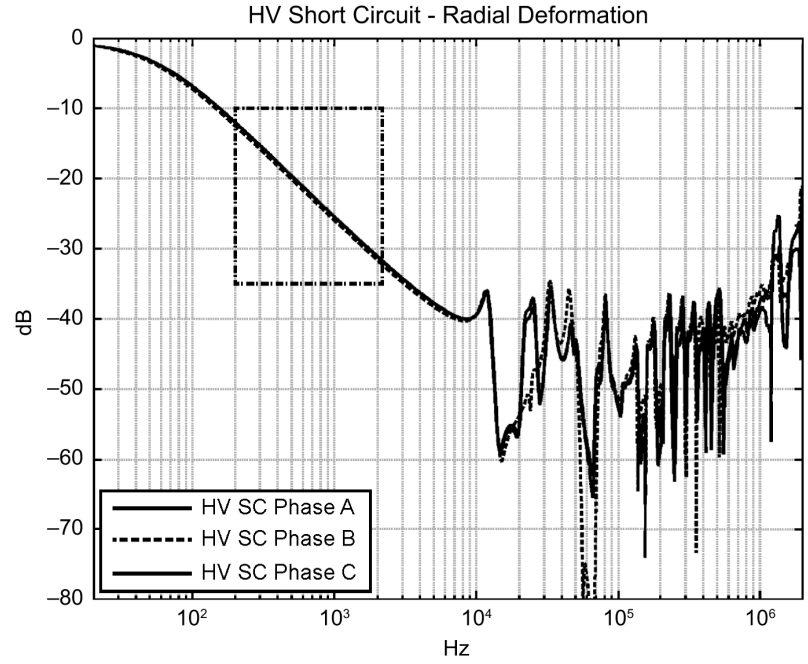


Figure 7—Radial movement response from HV short-circuit tests

7.5.1.2 Case study 2—Radial deformation

Figure 8 and Figure 9 were collected from a transformer that exhibited radial deformation on the LV winding. Phase C was most affected. A change/shift in the 50 kHz to 1 MHz frequency range in Figure 8 is obvious. Upon closer examination, Phase C is also slightly lower in the low frequency range in Figure 9. Both of these finding identify radial deformation.

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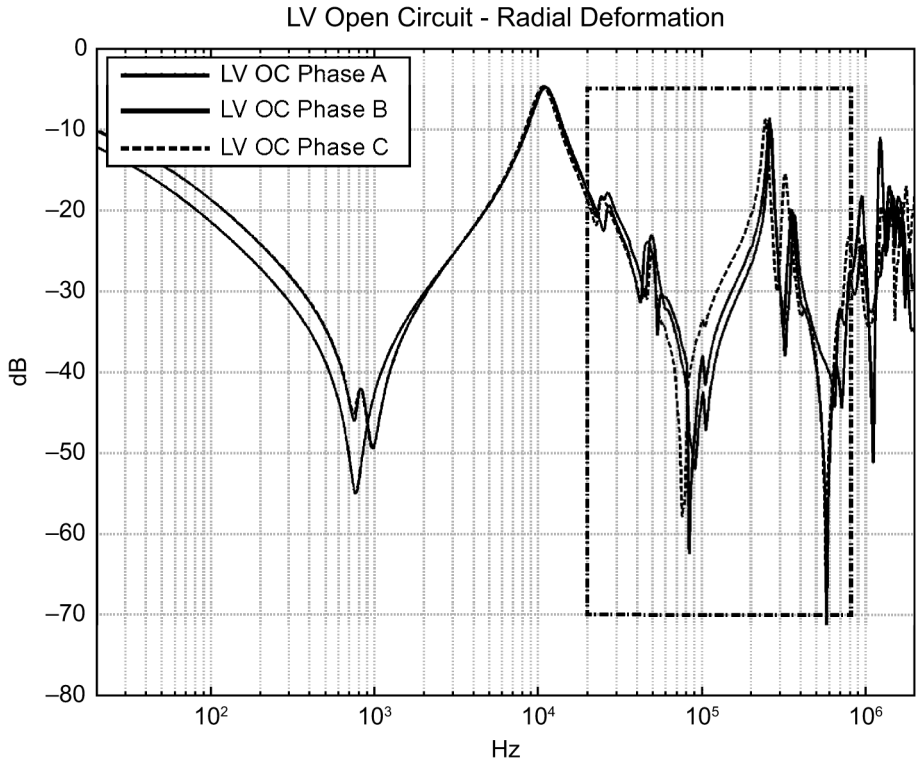


Figure 8—Radial movement response from HV short-circuit tests

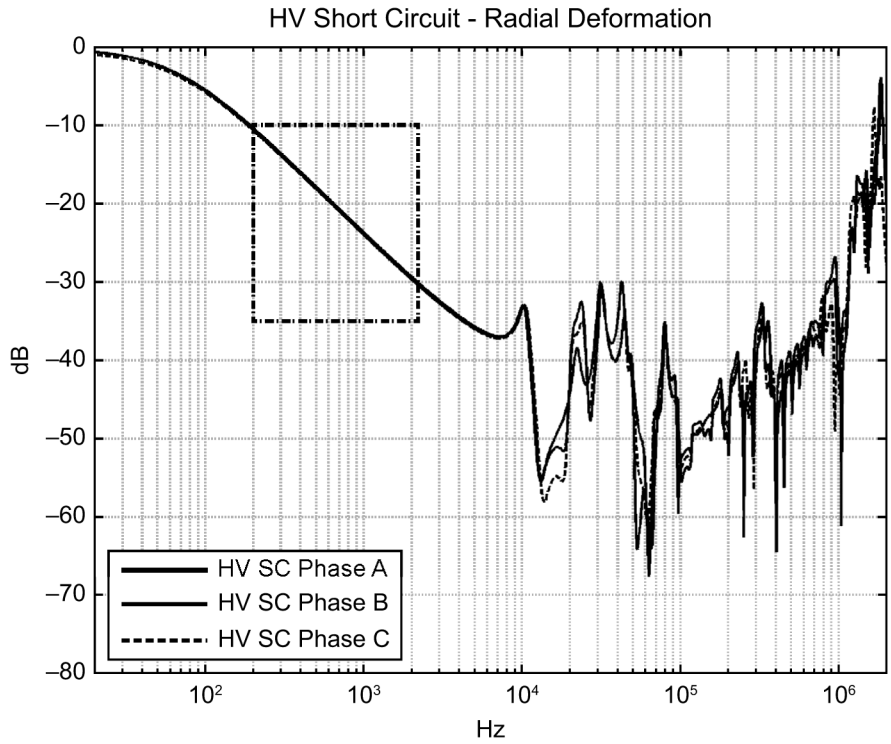


Figure 9—Radial movement response from HV short-circuit tests

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7.5.2 Axial winding elongation—"telescoping"

Axial winding movement includes two types of winding geometric changes. The winding is stretched or "telescoped" and then tightens due to a reduction in the windings radius. The geometric variations induced by this type of failure are complex and can lead to multiple resonances shifting across a broad frequency range.

Axial winding deformation affects the FRA measurements as shown in Table 6:

Table 6—Axial winding deformation

Frequency range	Axial winding deformation Assuming, no other failure modes exist:
20 Hz to 10 kHz	**Open circuit tests:** This region (core region) is generally unaffected during axial winding deformation. **Short circuit tests:** Results in a change in impedance. The FRA trace for the affected winding causes a difference between phases or previous results in the inductive roll-off portion.
5 kHz to 100 kHz	**Open circuit and short circuit tests:** Axial winding deformation is most obvious in this range. The bulk winding range can shift or produce new resonance peaks and valleys depending of the severity of the deformation. The changes will be greater for the affected winding, but it is still possible to have the effects transferred to the other winding(s).
50 kHz to 1 MHz	**Open circuit and short circuit tests:** Axial winding deformation can shift or produce new resonance peaks and valleys depending of the severity of the deformation. The changes will be greater for the affected winding, but it is still possible to have the effects transferred to the other winding(s).
> 1 MHz	**Open circuit and short circuit tests:** The response to axial winding deformation is unpredictable.

Typical "axial" winding elongation results are shown in Figure 10 and Figure 11.

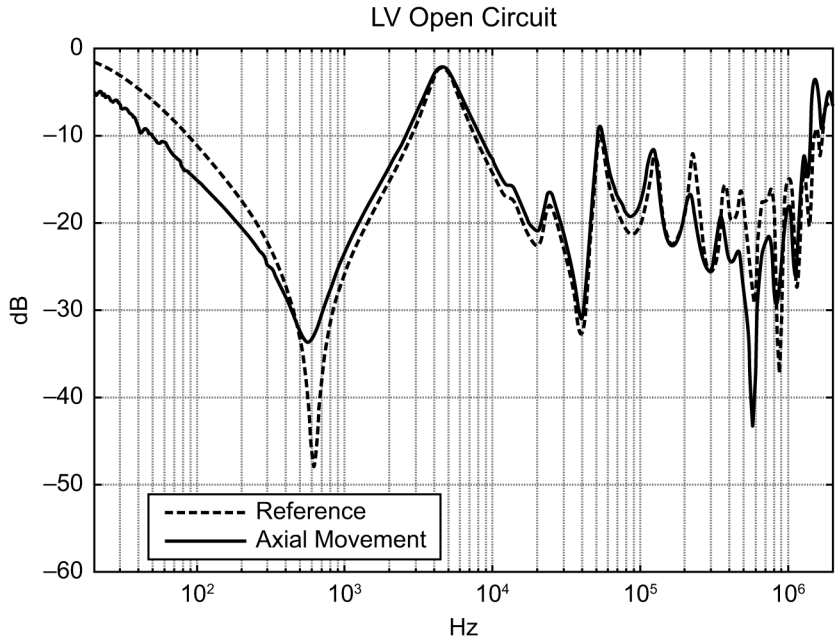


Figure 10—Axial movement from tertiary open-circuit test

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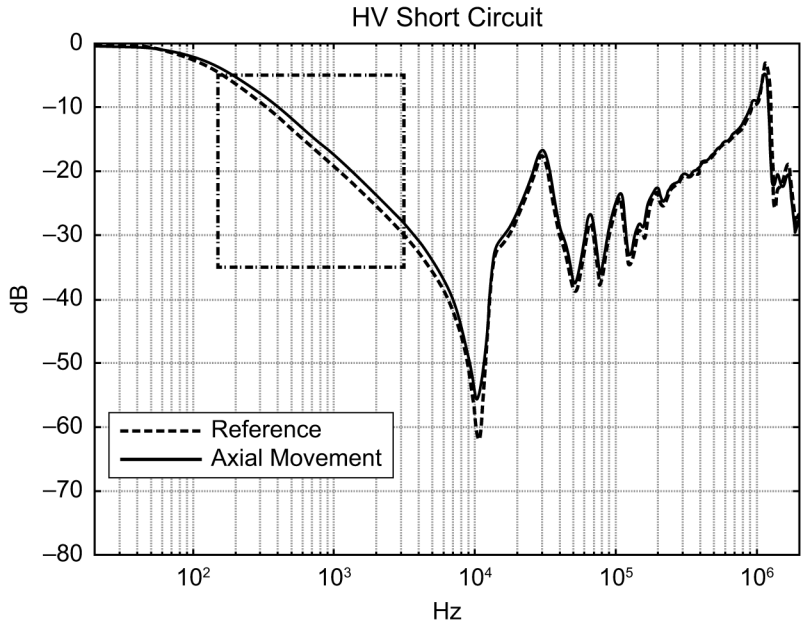


Figure 11—Axial movement response from HV short-circuit test

7.5.3 Overall bulk movement

These related failure modes describe the overall movement of windings or sections of windings. Considering a general movement of the winding, the causes could be due to a variety of reasons. Generally, this type of failure is used to describe the movement of the coils due to physical shock as a result of high current forces or transportation. Physical movement of the transformer could be due to shipping or seismic activity.

Bulk winding deformation affects the FRA measurements as shown in Table 7:

Table 7—Bulk winding deformation

Frequency range	Bulk winding deformation (assuming no other failure modes exist)
20 Hz to 10 kHz	**Open circuit tests:** This region (core region) is generally unaffected during bulk winding movement. **Short circuit tests:** This region is generally unaffected during bulk winding movement. All phases should be similar.
5 kHz to 100 kHz	**Open circuit and short circuit tests:** Bulk winding movement is most obvious in this range. Newly created resonance peaks or valleys are the key indicator. The bulk winding range can shift or produce new resonance peaks and valleys depending of the magnitude of the movement. The changes will be greater for the affected phase.
50 kHz to 1 MHz	**Open circuit and short circuit tests:** Generally, this range remains unaffected. However, changes to the CL capacitance can cause resonance shifts in the upper portion of this range.
> 1 MHz	**Open circuit and short circuit tests:** Changes to the low-voltage winding-to-ground insulation capacitance (CL) can cause resonance shifts.

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7.5.4 Core defects

Core defects failures cause changes to the core's magnetic circuit. Core defects can include burnt core laminations, shorted core laminations, multiple/unintentional core grounds, lost core ground, and joint dislocations. Details are as follows in Table 8.

Table 8—Core defects

Frequency range	Core defects (assuming no other failure modes exist)
20 Hz to 10 kHz	**Open circuit tests:** These types of failures will affect the lower frequency regions generally below 10 kHz. Core defects often change the primary core resonance shape. Less weight should be placed on shifting, because identifying core defects can sometimes be masked by the effects of core residual magnetization. If the open circuit core appears loaded, (i.e., looks like a short circuit response), this could indicate a core defect. **Short circuit tests:** This region is generally unaffected during bulk winding movement. All phases should be similar.
5 kHz to 100 kHz	**Open circuit and short circuit tests:** This range can shift or produce new resonance peaks and valleys.
50 kHz to 1 MHz	**Open circuit and short circuit tests:** Generally, this range remains unaffected. However, if the fault is due to a core ground issue, resonance shifts may appear in the upper portion of this range.
> 1 MHz	**Open circuit and short circuit tests:** If the fault is due to a core ground issue, resonance shifts may appear in this range.

Typical "core defect" results are shown in Figure 12 and Figure 13.

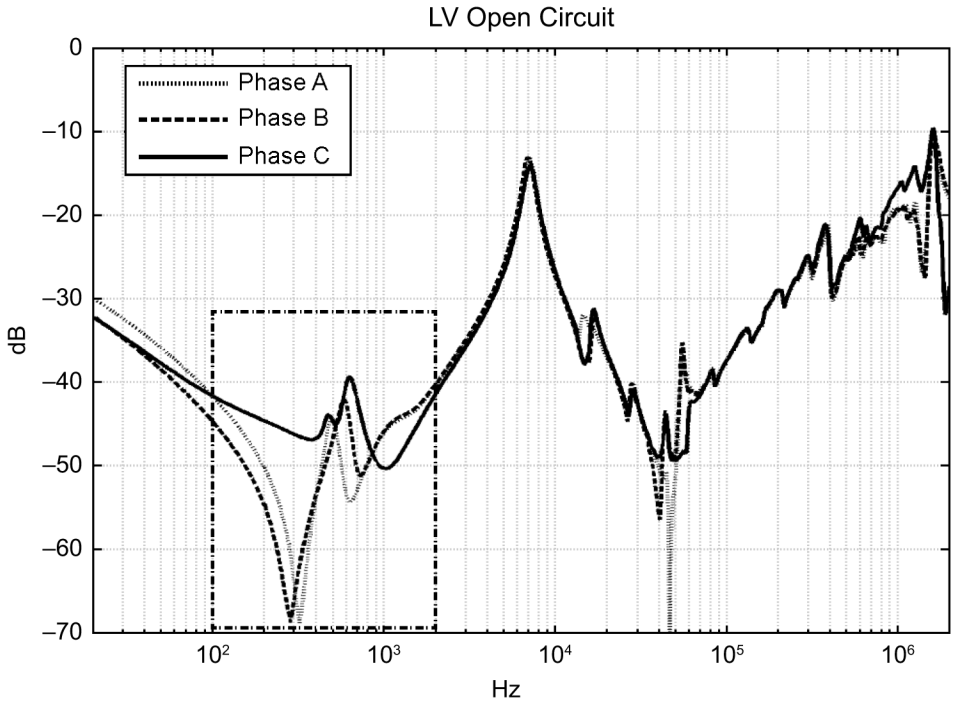


Figure 12—Core defect response from LV open-circuit test

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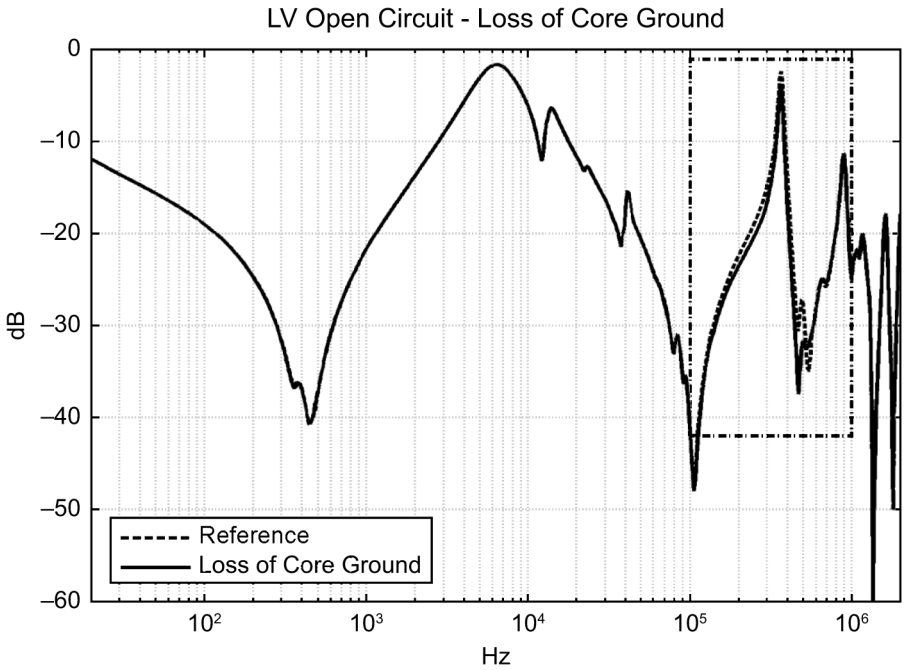


Figure 13—Loss of core ground from LV open-circuit test

7.5.5 Winding turn-to-turn short circuit

Turn-to-turn faults are arguably one of the easier failure modes that can be identified by the FRA test. Turn-to-turn short-circuits can occur between two neighboring turns or between phases. The short can be either a low impedance solid short or high resistance leakage path. These details are elaborated in Table 9.

Table 9—Winding turn-to-turn short circuit

Frequency range	Winding turn-to-turn short circuit Assuming, no other failure modes exist:
20 Hz to 10 kHz	**Open circuit tests:** The short circuit failure mode removes the effect of the core's reluctance from the open circuit FRA results. The FRA open circuit trace assumes a similar behavior as the short circuit test. The affected winding will show the greatest change. This failure mode will also affect the FRA responses for all other windings, but not as much. **Short circuit tests:** The results will not compare well against previous data or among phases. The affected winding is generally offset.
5 kHz to 100 kHz	**Open circuit and short circuit tests:** This range can shift or produce new resonance peaks and valleys. The changes will be greater for the affected phase.
50 kHz to 1 MHz	**Open circuit and short circuit tests:** This range can shift or produce new resonance peaks and valleys. The changes will be greater for the affected phase.
> 1 MHz	**Open circuit and short circuit tests:** This range can shift or produce new resonance peaks and valleys. The changes will be greater for the affected phase.

Typical "turn-to-turn short" results are shown in Figure 14 and Figure 15.

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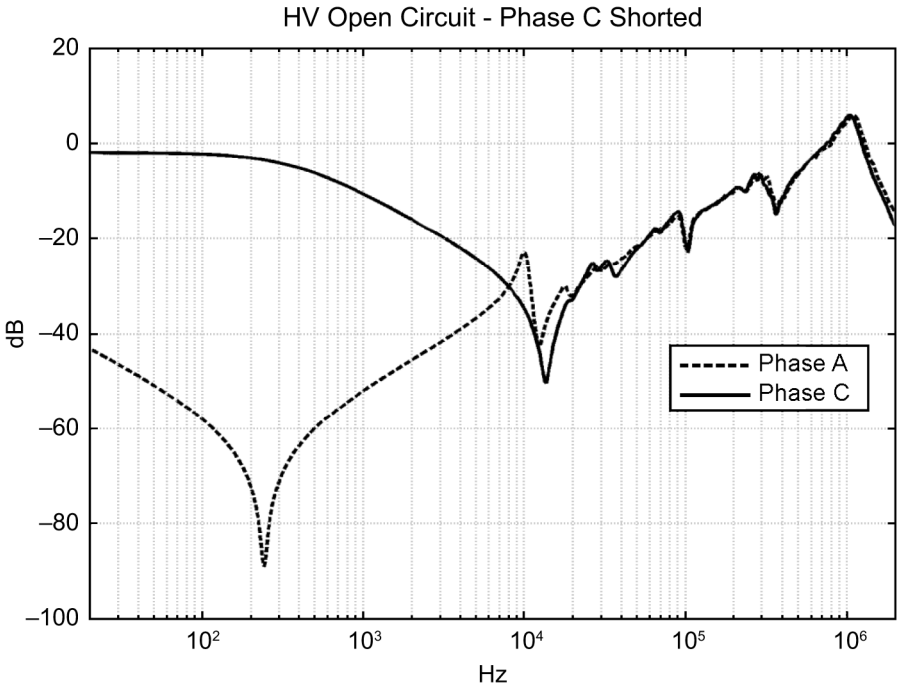


Figure 14—Turn-to-turn short response from HV open-circuit test

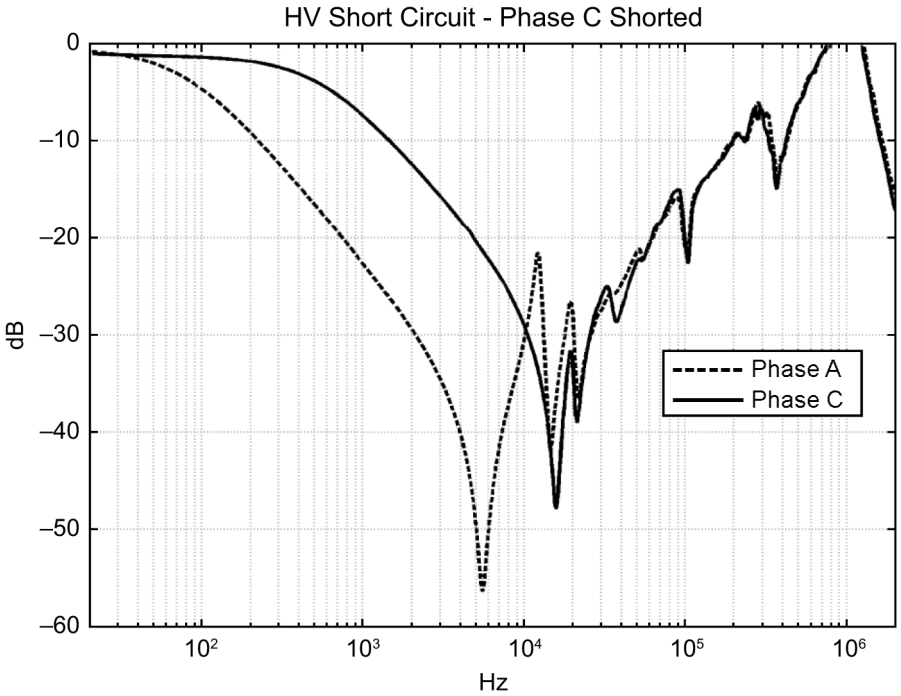


Figure 15—Turn-to-turn short response from HV short-circuit test

7.5.6 Residual magnetization

Though not necessarily a failure, residual magnetization within the core should be identified, so as not to be misinterpreted as an actual fault. Residual magnetization is the flux density that remains in the core steel.

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DC winding resistance testing, switching operations, and geomagnetic phenomena are sources of residual magnetism. Residual magnetization can be identified by the shifting of the low frequency core resonance to the right compared to the demagnetized results. Residual magnetization can be removed by demagnetizing the core, and should be conducted if there is concern about the condition of the core.

7.6 Modeling

Practically, there are two methods to obtain the frequency response characteristic of a transformer winding. The first is to make physical measurements on the physical transformer winding structure.

The second is to model the transformer analytically for example with an RLC network, and mathematically compute the desired response curves. The advantage of this is that all locations are mathematically accessible and it is possible to simulate "fault" conditions without physically damaging a transformer.

The challenge is to build a valid and accurate analytic model. As discussed in the FRA theory section, it is necessary to represent the complete three-phase transformer in order to simulate an accurate FRA response. A complete model of the windings can be constructed using a series of n-stage ladder networks such as shown in Figure 16 to cover the required frequency range.

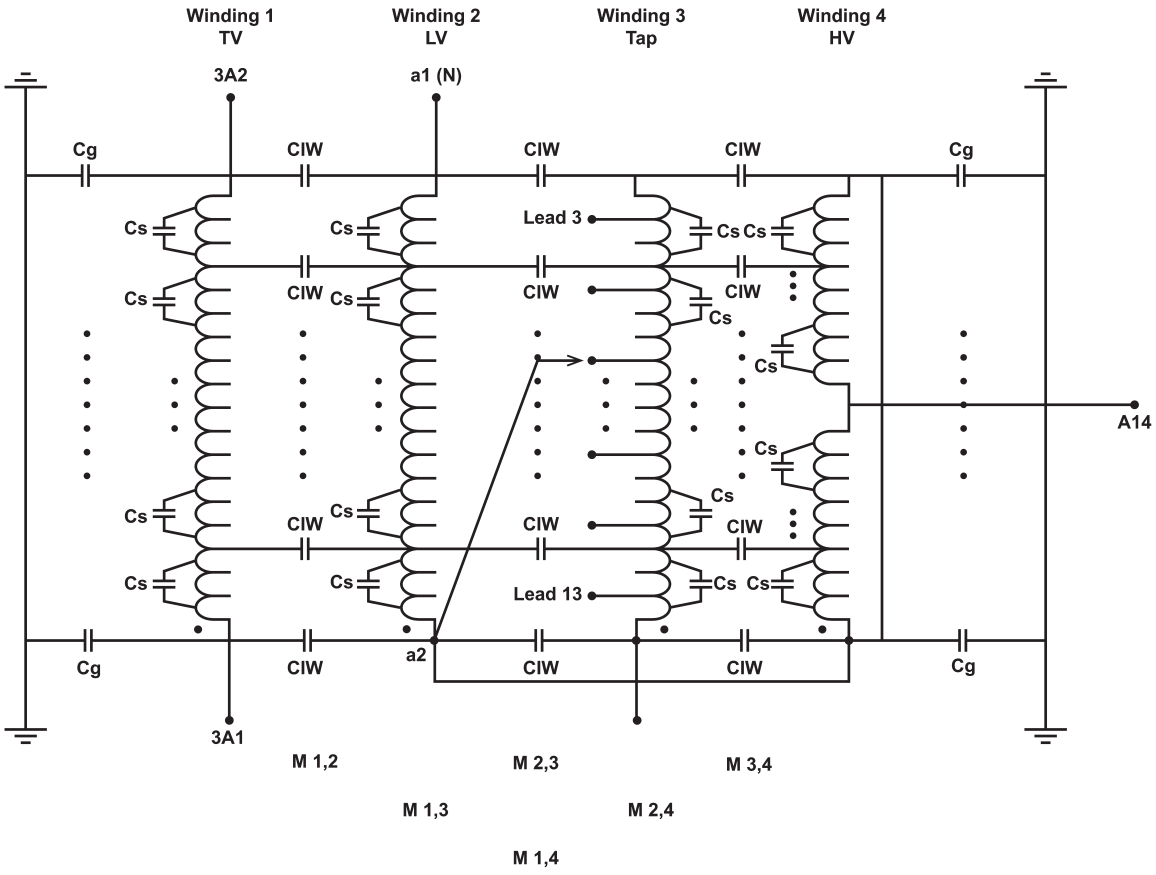


Figure 16—n stage ladder network of a transformer

The inductance matrix would include the core and the leakage inductances, whereas the capacitance matrix would include the shunt and series capacitances. Conductor loss and dielectric loss are represented as resistance, which is frequency-dependent. The effect of any three-phase connections is recreated by further

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extension of the model. The end result can be a very large model to be solved in the frequency domain, requiring lengthy computation, but with modern computers, this is no longer a problem. It has been shown, however, that it is feasible to make certain simplifications to the model and its component matrices without loss of accuracy in the critical regions of the response curve [B13].[12]

It is obviously important to understand the difference between the physical device and the mathematical model that is used. However, if an existing transformer is modeled that has had sufficient FRA measurements carried out both during and after its construction; it is possible to validate the model with a good degree of confidence [B10].

It is now possible to use the validated model to simulate particular physical changes, such as winding damage or movement, in order to see the changes to the frequency response. Sensitivity studies, effects of different connections, etc., can now be carried out to aid the use of the FRA diagnosis tool and improve the ability to make confident interpretation of response curve changes.

[12]The number in brackets correspond to those of the bibliography in Annex A.

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Annex A

(informative)

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Annex B

(informative)

FRA theory

Frequency response analysis (FRA) is a technique that is used to diagnose the condition, or more importantly the change of mechanical condition, of a transformer by analyzing the transformer winding's frequency characteristic. FRA provides internal diagnostic information using non-intrusive procedures.

The technique consists of measuring the FRA magnitude, see Clause 2 Definitions, e.g., admittance (the current at the response end of the winding divided by the voltage at the other input end of the winding) of the transformer winding largely comprised of capacitive and inductive elements. The measurement is performed over a wide range of frequencies and the result is compared with a reference "signature" or "fingerprint" result of the winding to make a diagnosis.

The measurements are generally made across the two terminals of a winding (across the winding) to derive the winding end-to-end frequency response in the form of magnitude against frequency as for the commonly used admittance is given by Equation (A.1).

$$A(f) = 20\log_{10}\left|\frac{\dot{V}_2(f)}{\dot{V}_1(f)}\right|, \text{dB} \tag{B.1}$$

Where

A is the amplitude calculated at a specific frequency f in dB
$\dot{V}_1$ is the source voltage applied at one end of winding
$\dot{V}_2$ is the measured quantity at the other end of the winding

In theory, the second measurement $\dot{V}_2$is a replica of the current response at one end of winding to the injected voltage at the other end of the winding. Therefore, the winding end-to-end response can be regarded as the "virtual" self-admittance of the winding.

The measured response is usually shown graphically by plotting the logarithmic amplitude ratio of the output voltage-input voltage in dB (y axis) against the frequency (x axis). The frequency scale can be logarithmic or linear. Both are used, although the logarithmic often shows the complete frequency range more clearly. The linear scale is useful for looking at discrete frequency bands and to compare small differences at particular frequencies. Typical responses of the LV windings of a transformer (see [B16]) are shown below using both logarithmic, Figure B.1, and linear scales, Figure B.2.

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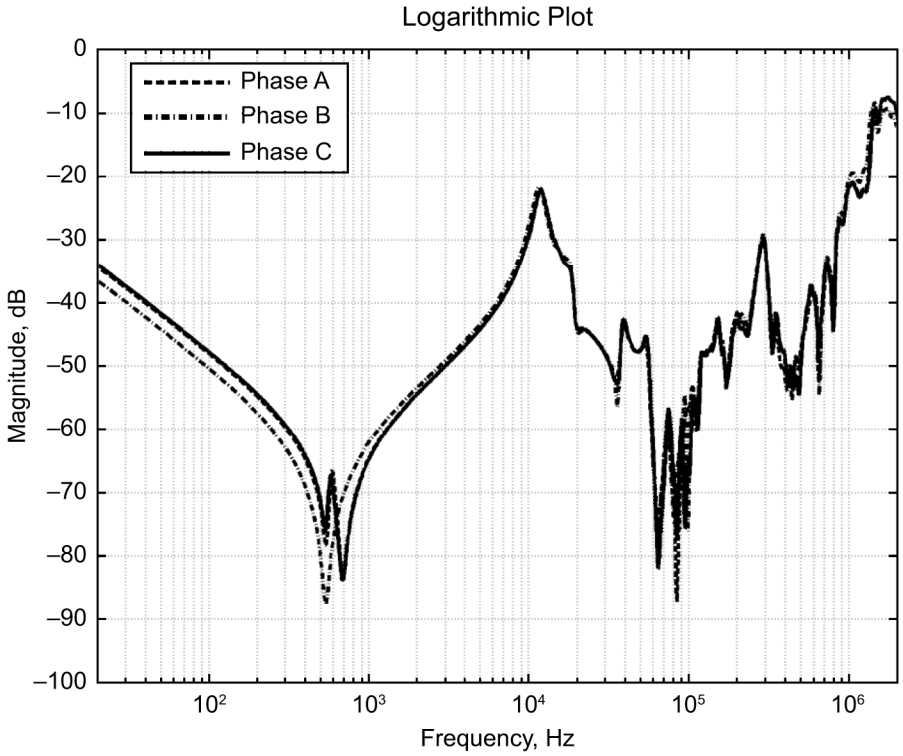


Figure B.1—Logarithmic frequency response

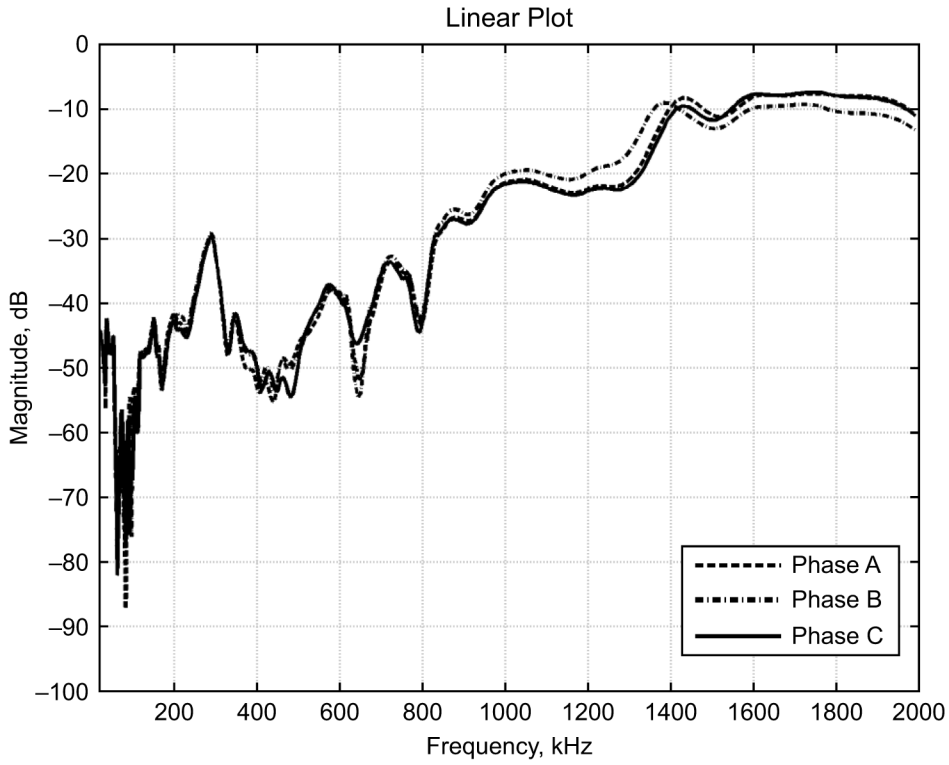


Figure B.2—Linear frequency response

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As can be seen, the frequency response of a transformer winding (often called the FRA response curve) is quite complex and consists of decreasing and increasing magnitude (in dB) with respect to frequency. The various resonances (maxima) and anti-resonances (minima) are determined by the electrical characteristics of the transformer winding. These characteristics can be represented by the transformer equivalent circuit and would include the elements of resistance, inductance, and capacitance. The inductance and capacitance values in this equivalent circuit are determined by winding structure and geometry, and insulation structure and clearance, and the resistance is contributed by conductive loss and dielectric loss.

In order to better understand and interpret possible changes to the FRA response, it is useful to understand which elements are responsible for the various regions and shapes of the response curve. This can be realized by examining the individual responses of the various elements and their combinations.

Taking the simplest representation of a single winding as a lumped circuit elements network, and initially ignoring the inductive and capacitive couplings between windings, the end-to-end frequency response of a winding is dependent on how the elements in this network behave together at different frequencies.

At low frequencies, a transformer winding behaves as an inductive element, and the end-to-end FRA response follows a falling magnitude trend across the frequency range with a linearly decreasing slope of approximately –20 dB per decade. A higher inductance causes the magnitude to decrease. Power transformers with higher voltage and larger power rating usually have larger negative response magnitudes at low frequencies. Effectively there are two inductance components affecting the frequency response; one is the core magnetizing inductance, and the other is the leakage inductance of the winding. Each affects the response in a different frequency region. The former affects the FRA response in the lower frequency region up to ~100 Hz while the latter influences the FRA response at higher frequencies.

At high frequencies, a transformer winding behaves as a capacitive element, and the end-to-end FRA response follows a rising magnitude trend across the frequency range with a linearly increasing slope of approximately 20 dB per decade. A higher capacitance causes the magnitude to increase. Power transformers having both higher voltage and larger power rating usually have smaller negative response magnitudes at high frequencies.

The combination of winding inductance and winding series capacitance results in paralleled inductance and capacitance. LC in parallel will produce parallel anti-resonance at a certain frequency, blocking the signal at that particular frequency. This consequently produces a local anti-resonance in the magnitude response at that particularly frequency.

In the case that the winding series capacitance is relatively small (a single layer winding or a plain disc winding as an example), the shunt capacitance becomes significant. The combination of winding inductance and shunt capacitance will result in inductance and capacitance in series and produce series resonance. The simplest representation of LC in series is a T-connection where the shunt capacitance is connected in the middle of the two halves of the winding inductance. The end-to-end FRA response of a T–connected LC network shows a series resonance at a certain frequency, amplifying the signal at that particular frequency. This consequently produces a resonance at that particularly frequency.

As shown in Figure B.3, the effect of resistance on the end-to-end FRA response is to attenuate (smooth) the sharpness of the resonances and the anti-resonances. Conductor loss and dielectric loss are represented as resistance, which is frequency-dependent. In the equivalent circuit of a transformer winding, these are either connected in series with the inductance or connected in parallel with the capacitance.

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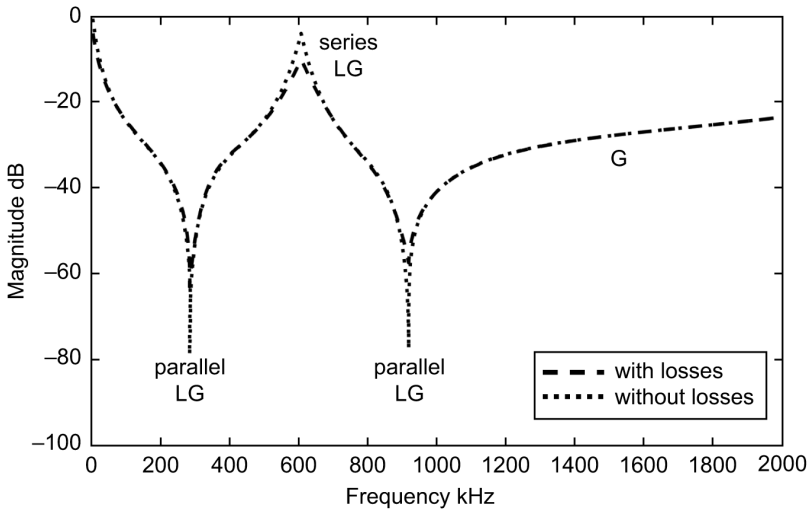


Figure B.3—Parallel and series resonant point

However, even in a single winding, these basic LC components are produced by mutual coupling between turns and parts of a winding, effectively resulting in a network of multiple lumped parameters. Figure B.4 and Figure B.5 show first, the inclusion of mutual coupling and second, the inclusion of losses.

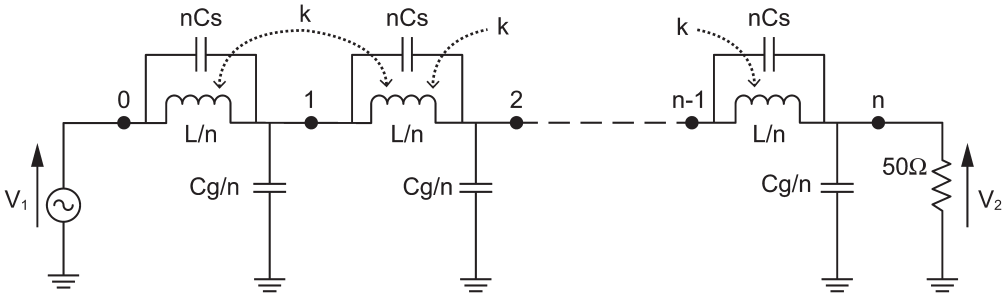


Figure B.4—n-stage lumped network with mutual coupling

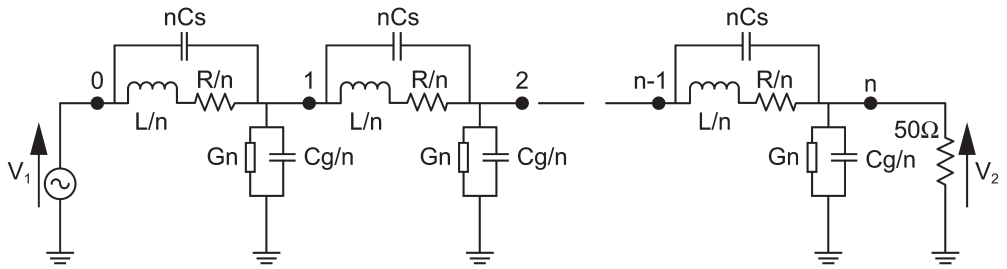


Figure B.5—n-stage lumped ladder network with losses

Thus, the winding equivalent circuit now comprises multiple distributed parameters, and the frequencies of the resonances are determined primarily by the winding leakage inductance L, the shunt capacitance Cg, and the winding series capacitance Cs. There are two extreme cases.

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In the one case, when the winding series capacitance is small enough to be neglected (where Cs tends to zero), there will be resonances with even frequency intervals. On the other hand, when the winding series capacitance is extremely large (where Cg tends to zero), there will be no resonance at all. With low Cs, the response begins with a flat magnitude trend and resonances at intervals of frequencies determined by the L, Cg, and Cs values and then followed by a decreasing inductive trend. An anti-resonance appears at a frequency determined by L and Cs, which is followed by the increasing capacitive trend. As Cs is increased, some of the resonances diminish and the anti-resonance appears at a lower frequency. Diagrams that explain this paragraph better can be found in reference [B18].

The relative proportions of the series capacitance (Cs) and the ground capacitance (Cg) are thus significant in determining the FRA response for a specific winding structure. They determine not only which winding type has the higher magnitude, but also the shapes and the position of the resonances and anti-resonances and whether these appear at lower or higher frequencies.

In a “practical” transformer, there are at least two windings per phase and the interaction between these two windings needs to be considered. In general, this interaction can be described by inductive and/or capacitive coupling. There are also interactions between windings of different phases due to their electrical connections, such as for a delta, or the sharing of the same neutrals. Windings not under test and not electrically connected to the tested winding, will also have an impact on the frequency response through mutual inductive and capacitive couplings.

Summary of effects

Knowledge of the above effects and the response “shapes” for different windings and combinations of windings is useful in making diagnostic interpretation of changes to the FRA response curves.

Figure B.6 and [B16] shows typical frequency responses for the HV windings of an autotransformer taken from 10 Hz up to 10 MHz and shown on the logarithmic scale. The frequency range can be divided into 4 regions depending on the dominant influence of the various electrical properties of the transformer with frequency.

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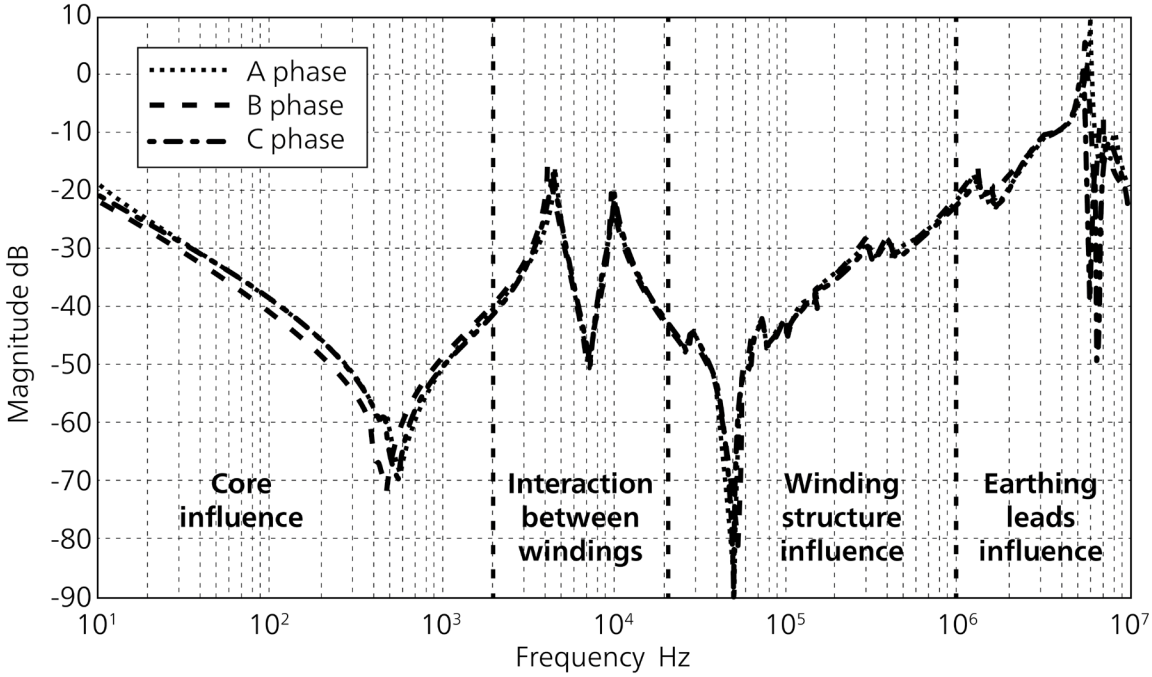


Figure B.6—Frequency region responses

The following frequency regions described here are typical but not exact, and they would be slightly varied depending on transformer design and arrangement.

a) Region 1—Core effect

At low frequency (below 2 kHz), the frequency response begins with the decreasing magnitude of −20 dB/decade. This is due to the magnetizing inductance of the core. This is then followed by a minimum which occurs due to the series resonance between the magnetizing inductance of the core and the total capacitance of the transformer. In this example, the B phase is clearly different from the other two phases due to the different magnetic reluctance path of the middle leg of a three-limb core.

b) Region 2—Interaction between windings

Between 2 kHz to 20 kHz, the frequency response is influenced by the interaction between the windings as well as by how the windings and neutrals are connected and terminated (open/closed delta; floating/grounded).

c) Region 3—Effect of winding structure of the winding under test

From 20 kHz up to 1 MHz, the winding structure will heavily influence the frequency response. In this example, the FRA plot for this frequency range has less resonances and anti-resonance and a mainly capacitive raising trend, due to the high series capacitance of the HV winding.

d) Region 4—Effect of leads of taps and earthing leads

Beyond the frequency of 1 MHz, the trend of the frequency response is irregular and complex, influenced by the tap leads and the measurement earthing leads.

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